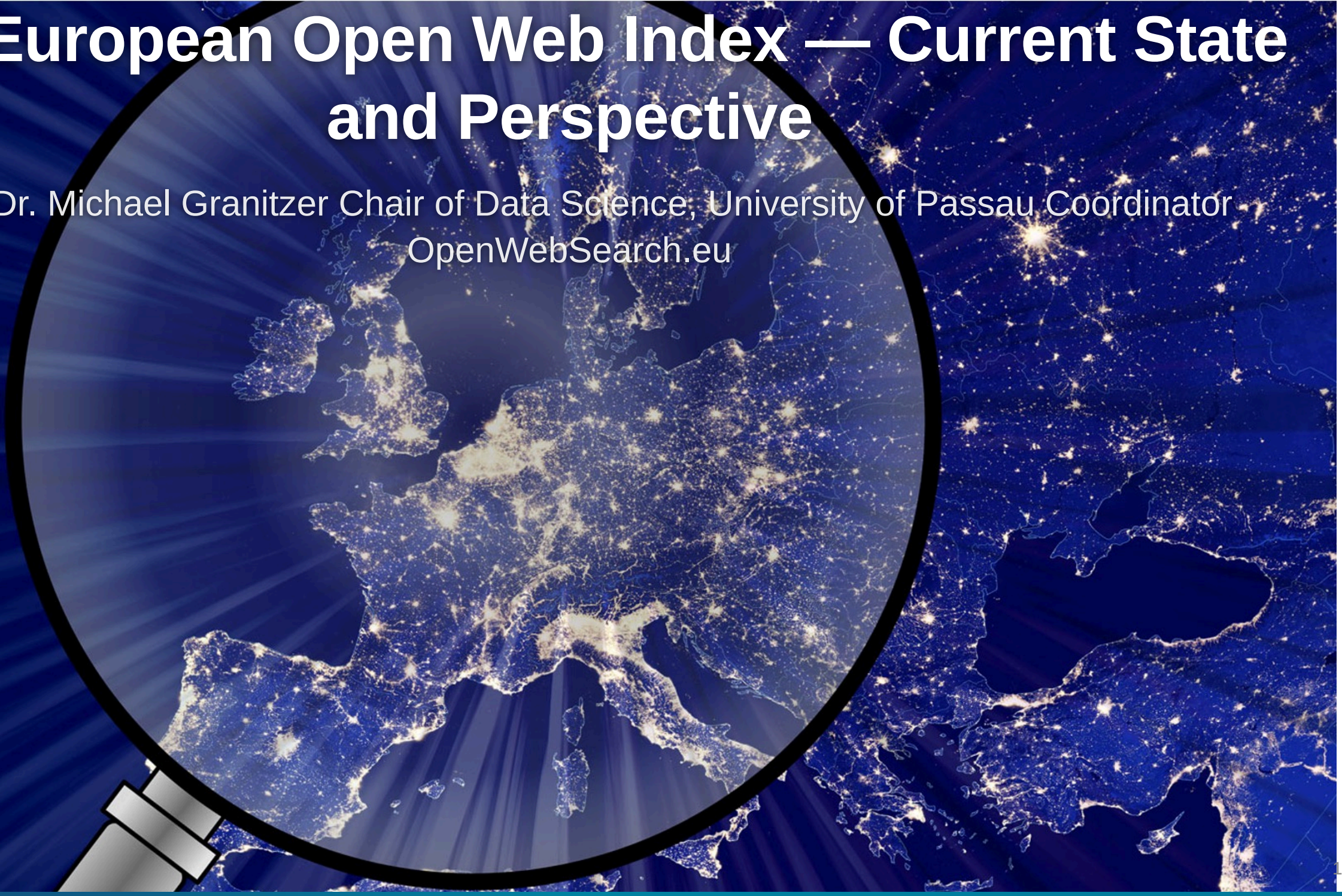


# The European Open Web Index — Current State and Perspective

Prof. Dr. Michael Granitzer Chair of Data Science, University of Passau Coordinator  
[OpenWebSearch.eu](http://OpenWebSearch.eu)



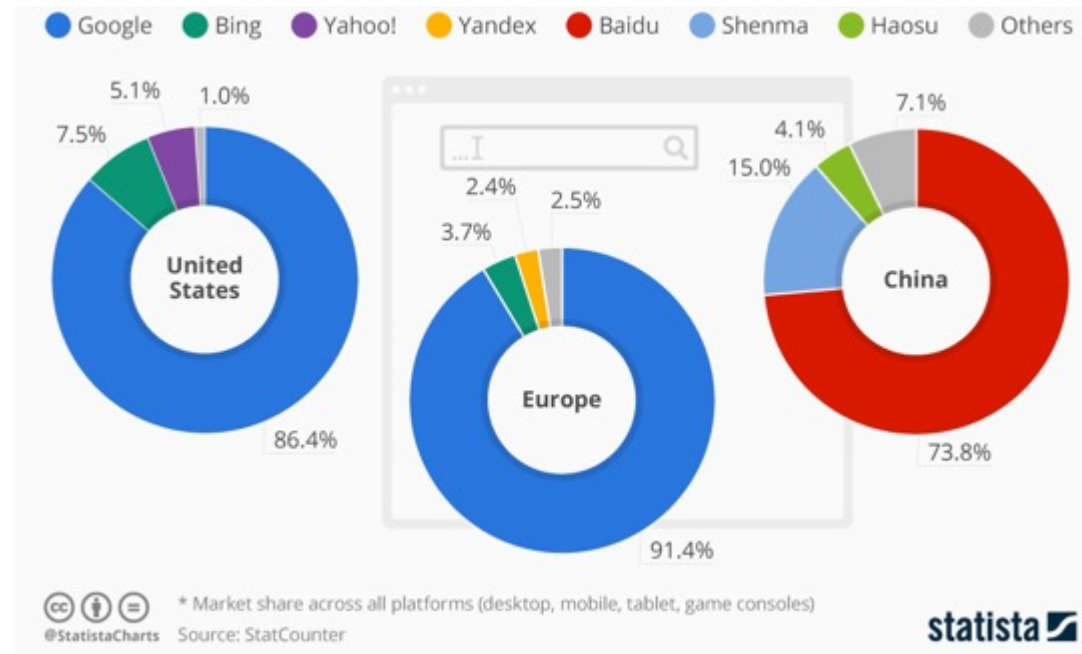


# Web Search Today

A critical resource managed as an oligopoly

## Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



# But does it matter? Yes — on four dimensions

## User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

## Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

## Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

## AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

Being able to navigate the Web is as fundamental as GPS or the power grid



# Challenges

## THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

### VOLUME

**1.15 Billion**

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

**Top 0.1%–1% of sites**

may account for roughly half of all pages  
heavy-tail intuition from large web graph studies

**~ 400 Billion**

documents in Google's search index  
2020 figure cited in US v. Google testimony

*Sites ≠ domains ≠ pages; use exact counts cautiously*

### VELOCITY

**+8.6 M / month**

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

**~ 3.2 / second**

same monthly net change, averaged across December

**149 ZB (global)**

all digital data created / copied worldwide in 2024  
IDC / Statista global datasphere estimate

*The web changes constantly; crawling is continuous, not one-off*

### VARIETY

**150+ languages**

observed across websites

W3Techs usage survey

**249 CMS Platforms**

CMS platforms identified in the HTTP Archive dataset  
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

**HTML · PDF · JSON-LD · media · APIs**

content types to parse and process

*HTML · PDFs · JS apps · schemas · media · APIs*

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·  
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista



- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties



# Challenges

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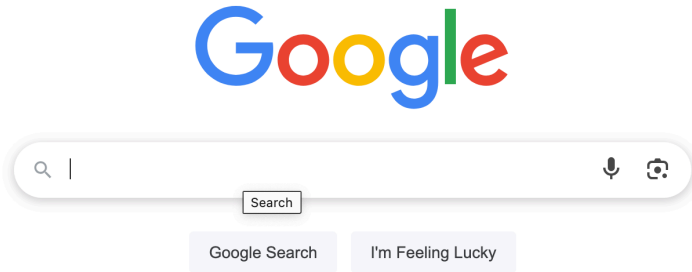


- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

High upfront costs to tap the Web as resources — especially small innovators and researchers are left behind

## Reduced user choice

Single source of our information: We would not be happy with only one European newspaper



Newspaper map: [reddit.com/r/MapPorn/comments/1c70v2q/newspapers\\_of\\_record\\_or\\_note\\_in\\_europe/](https://reddit.com/r/MapPorn/comments/1c70v2q/newspapers_of_record_or_note_in_europe/)

## Search Engine Manipulation Effect

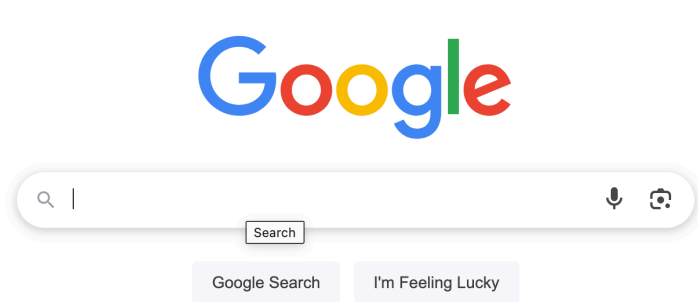
- Users locked into a single information gatekeeper
- **Search Engine Manipulation Effect (SEME)**: biased search rankings can shift voting preferences of undecided voters by **20% or more** — up to **72%** in some demographics — while users remain largely unaware of the manipulation





## Reduced user choice

Single source of our information: We would not be happy with only one European newspaper



VS.



Newspaper map: [reddit.com/r/MapPorn/comments/1c70v2q/newspapers\\_of\\_record\\_or\\_note\\_in\\_europe/](https://www.reddit.com/r/MapPorn/comments/1c70v2q/newspapers_of_record_or_note_in_europe/)

## Search Engine Manipulation Effect

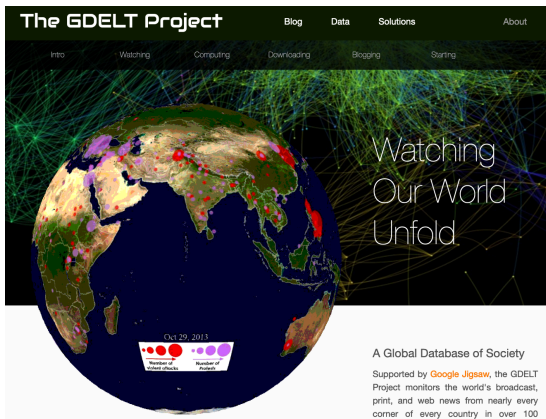
- Users locked into a single information gatekeeper
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# The web is the richest information source

The Web is the richest information source, but we are blind on what is there

- Companies, products, scientific information, events, geo-location data
- 60% of web pages contain structured data in form of Microdata / Microformats
  - Products, FAQ, Places, Shops
  - Job postings, Reviews, Events



Samsung GARMIN Fossil Apple Fitbit Price Reviews

Sponsored :

|                                                                                                                                                                                                                           |                                                                                                                                                                                                                         |                                                                                                                                                                                                                                             |                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Garmin VENU 2<br>Venu 2 -...<br>€222.22<br>Aika Juwelliere...<br>+€5.95 shipping<br>12 days max ·<br>Pedometer, Sle...<br>By Google | <br>Zeus - Smartwatch  ...<br>€89.95<br>StahlGear<br>Free shipping<br>★★★★★ (146)<br>10 days max ·<br>Pedometer, Sle...<br>By Klarna | <br>Gard Pro Health Smartwatch 2+...<br>€79.99<br>Gard Pro Deuts...<br>Free shipping<br>★★★★★ (6k+)<br>Pedometer, Sleep Tracker, Calori...<br>By Google | <br>Zeus - Smartwatch  ...<br>€89.95<br>StahlGear<br>Free shipping<br>★★★★★ (146)<br>Pedometer, Sleep Tracker, Calori...<br>By Klarna |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Places



**Kaufland Passau**  
4,3 ★★★★★ (2,8K) · Supermarket  
Neuburger Str. 128 · 0851 966350  
**Open** · Closes 8 pm  
In-store shopping · [Website](#)



**Rewe**  
4,1 ★★★★★ (1,3K) · Supermarket  
Nibelungenpl. 5 · 0851 4901046  
**Open** · Closes 8 pm  
In-store shopping · [Website](#)



**EDEKA Schwaiberg**  
4,2 ★★★★★ (120) · Supermarket  
Nikolastraße 2 · 0851 20091420  
**Open** · Closes 8 pm  
[Website](#)



# Societal Challenges which require a grip on the Web

## Misinformation / Disinformation

- Monopolistic ranking creates single points of failure for information quality
- Lack of index transparency prevents independent auditing and cross-language disinfo detection
- Open indices enable reproducible fact-checking pipelines

## Defence & Security

- European OSINT depends on US-controlled indices — strategic vulnerability in crisis scenarios
- Adversarial SEO ("search engine poisoning") is a documented hybrid warfare tactic
- No sovereign search infrastructure for independent web monitoring

## Guidance & Algorithmic Bias

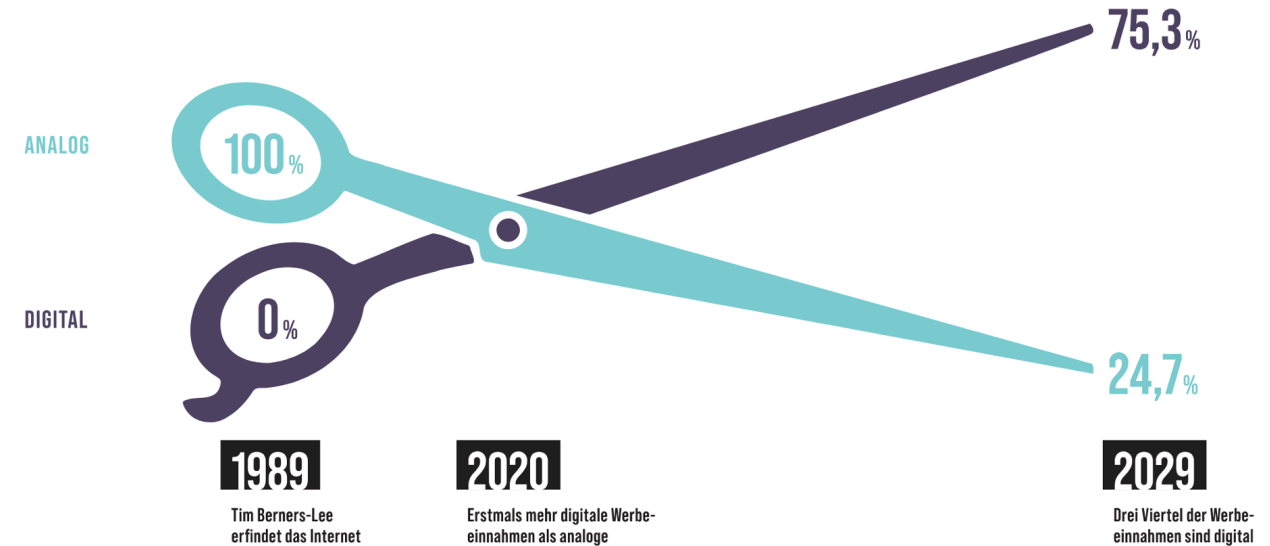
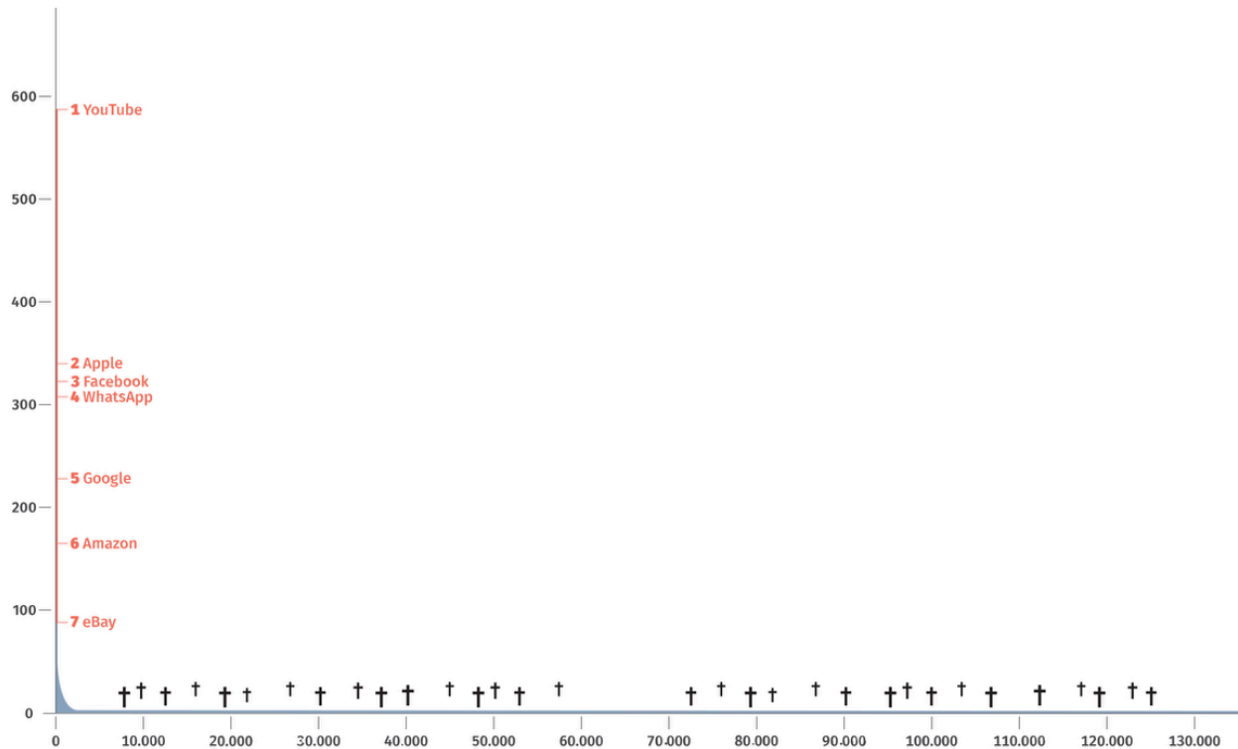
- Search autocompletions and knowledge panels shape health, legal and consumer decisions
- Filter bubbles create asymmetric information through personalization
- Low-income users disproportionately affected by low-quality results

## Digital & Information Literacy

- Most users cannot distinguish ads from organic results — "trust by ranking" effect
- Algorithmic literacy remains low even among university students
- Transparent, inspectable search infrastructure is a prerequisite for teaching information literacy effectively

# Europe is not present in the web

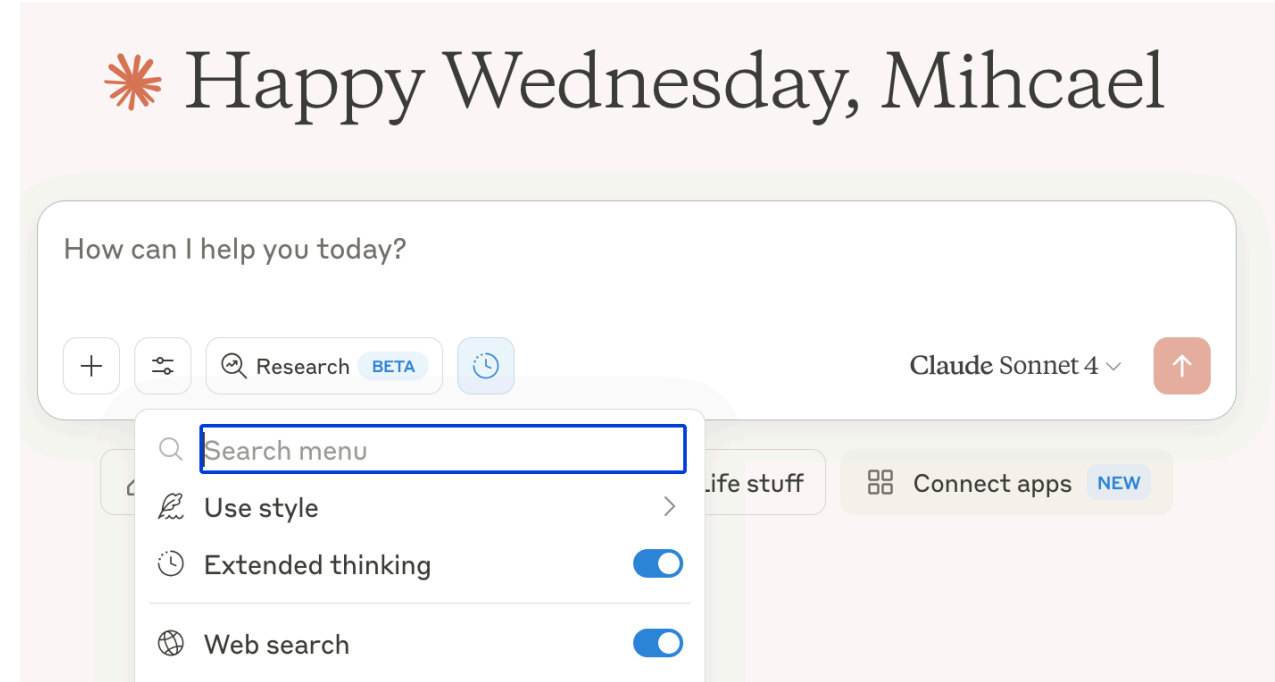
- Advertising revenue consolidated by US gatekeepers
- European content, languages, and perspectives underrepresented in search results
- No European alternative for web-scale data access





# The web drives innovation and AI

- Web data is essential for training LLMs and VLMs
- Retrieval Augmented Generation (RAG) requires web-scale document collections
- Agentic AI systems rely on web search as a core tool
- Web search remains an integral part of modern AI pipelines



Hey, Michael. Ready to dive in?

Ask anything

+

Tools

Create an image

Search the web

Write or code

Run deep research

🔊

⬆

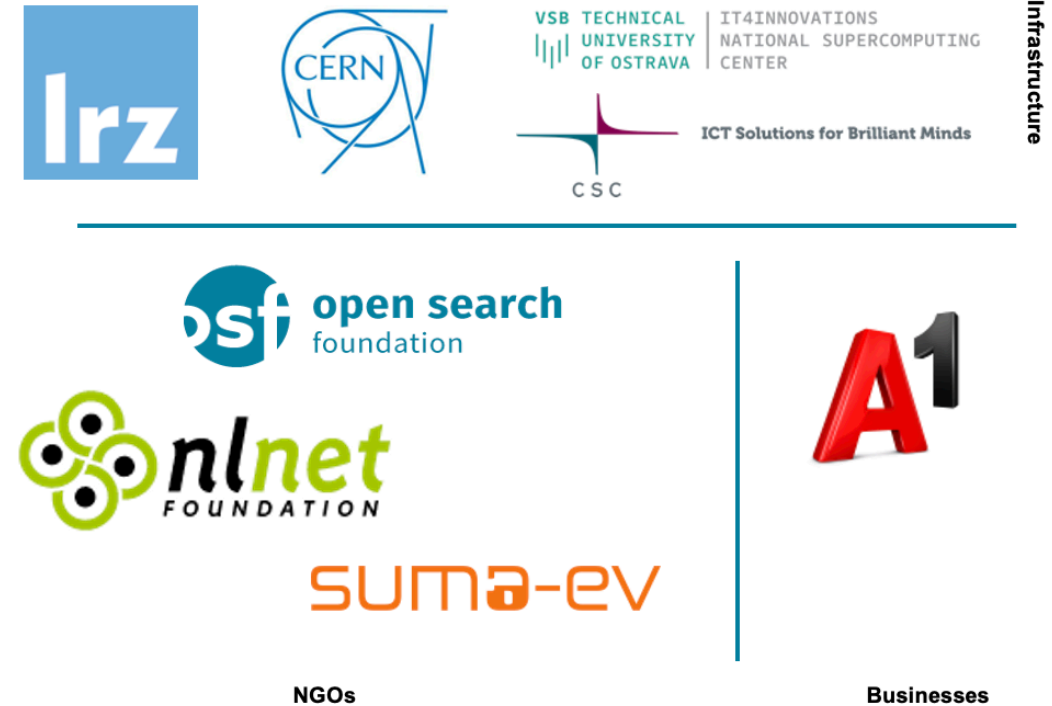
# OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

## Four Objectives

- 1. Open Technology Stack**  
Open-source pipelines for crawling, preprocessing, indexing
- 2. Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
- 3. Added Value Services**  
Dashboard, tools, data access APIs
- 4. Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

## 14 Core Partners

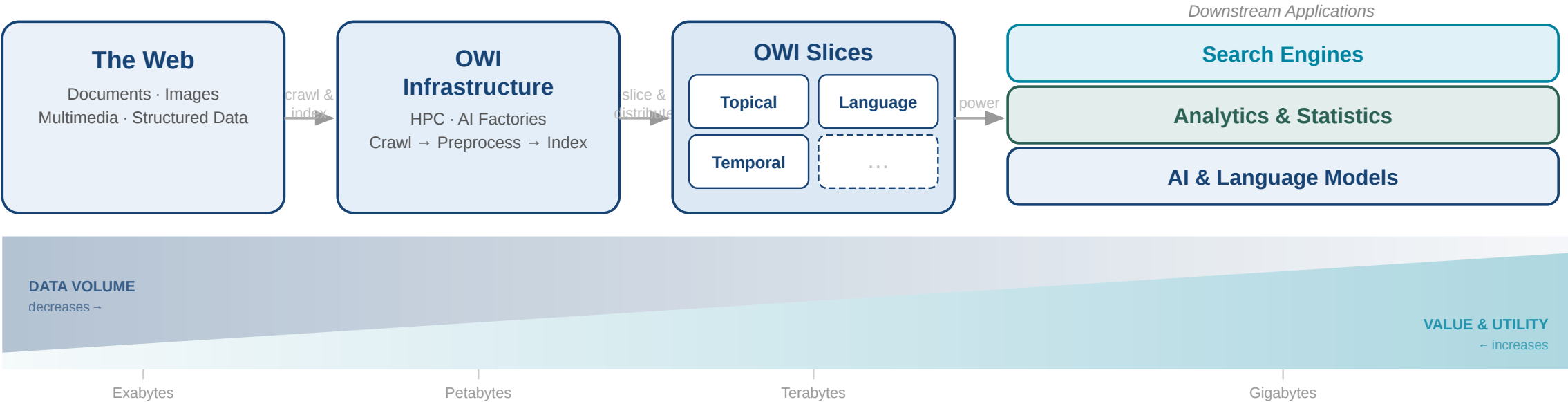


Plus 16 financially supported third-party projects (FSTPss)

# The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

**Our Proposition:** A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe

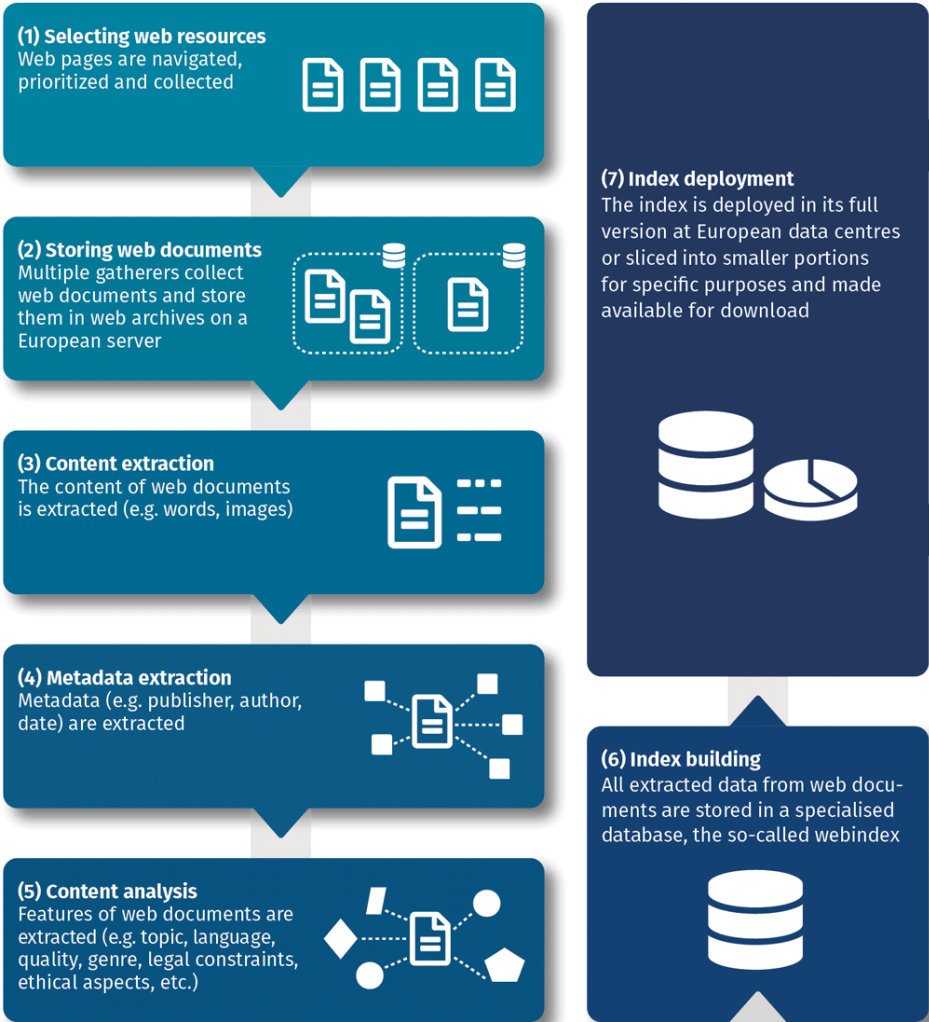


Granitzer, Michael, et al. "Impact and development of an Open Web Index for open web search." *Journal of the Association for Information Science and Technology* (2023).

# Conceptual Workflow

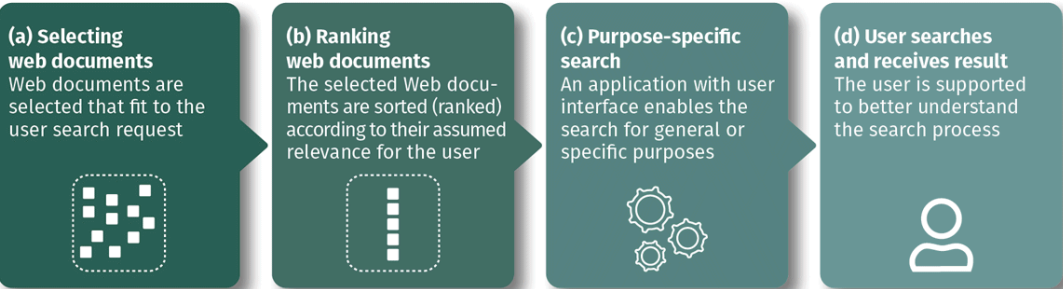
## Index Generation

Web resources are selected and retrieved, their content and metadata are analysed, and all data stored in the index database.



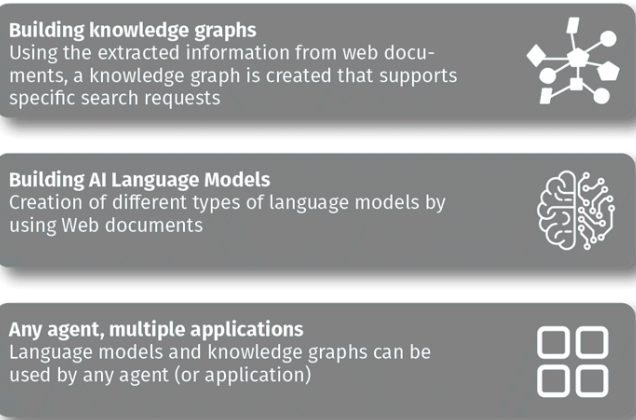
## Search Applications

A user search request will be answered by a search application that makes use of the open web index.



## Data Products

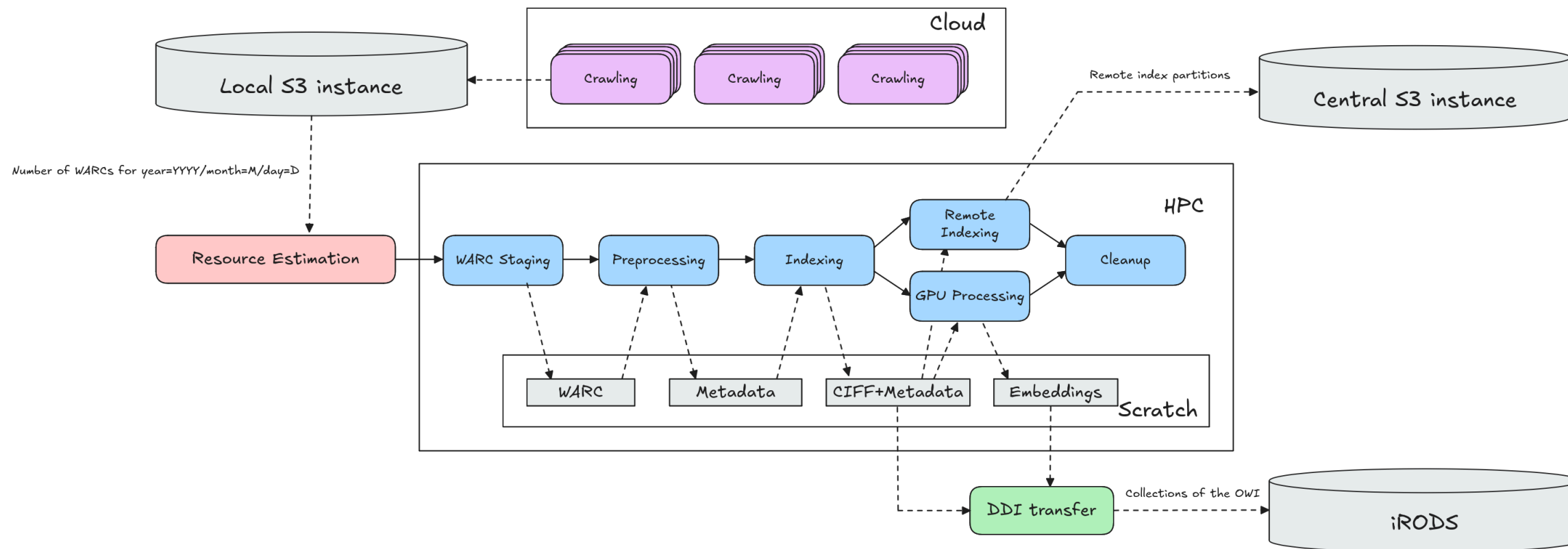
Knowledge representation models will be created using the open web index, in order to be used by any agent and for many applications







# Daily Workflow Pipeline



## Infrastructure (2026)

- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

## Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement





# The OpenWebSearch.eu Pipeline — Overview

## Step 1 — Crawling

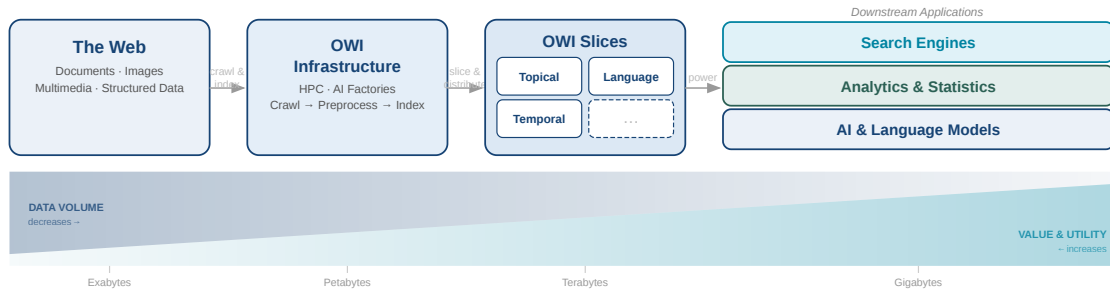
- 100 M URLs / day, 25 machines in 3–4 DCs
- European domain focus
- Respects robots.txt, Gen AI opt-out
- Full transparency via Dashboard

## Step 2 — Preprocessing (Resilipipe)

- HTML → clean text (ratio 1:5)
- Metadata: JSON-LD, Microdata, OpenGraph
- Semantic enrichment: language, topics, NER, geo-locations
- Structured data from 60 % of pages

## Step 3 — Index & Distribution

- Sparse full-text index (CIBF / Lucene)
- Feature index: link structure, style, genre
- Dense index: Jina V3 embeddings
- 810+ daily datasets via Owilix / Dashboard



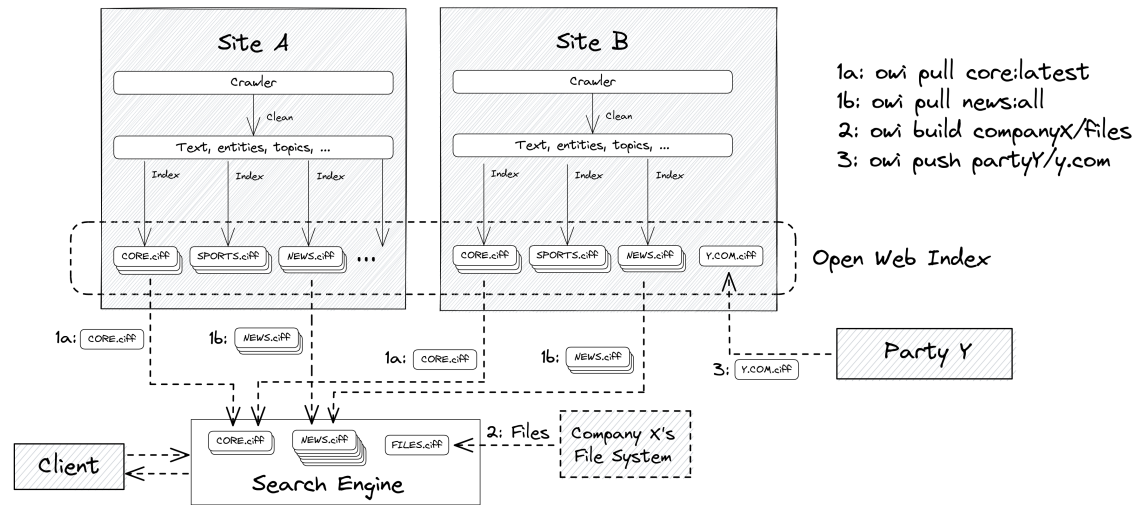
## Scale (as of early 2026)

- **25 VMs** across 3–5 data centres (IT4I, CSC, LRZ, CERN) running 24/7
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index in **810+** public datasets
- Factor **20 cheaper** than commercial indices

## Architecture principles

- Federated storage (iRODS + S3), cross-DC workflow via LEXIS
- Client-side SQL access via Owilix CLI
- Open format: CIFF sparse index + Parquet files
- Dense embeddings (Jina V3) available on select slices

# Architecture & Scale



## Four Cluster Tiers per Data Center

- **CCT** — OWLer crawler (StormCrawler-based); produces WARC files into S3
- **CFT** — singleton ScyllaDB URL frontier at CSC; coordinates all crawl sites
- **PET** — Resilipipe (Python multiprocessing + Resiliparse): WARC → Parquet
- **IST** — Open Web Indexer (Apache Spark): Parquet → CIFF + remote index

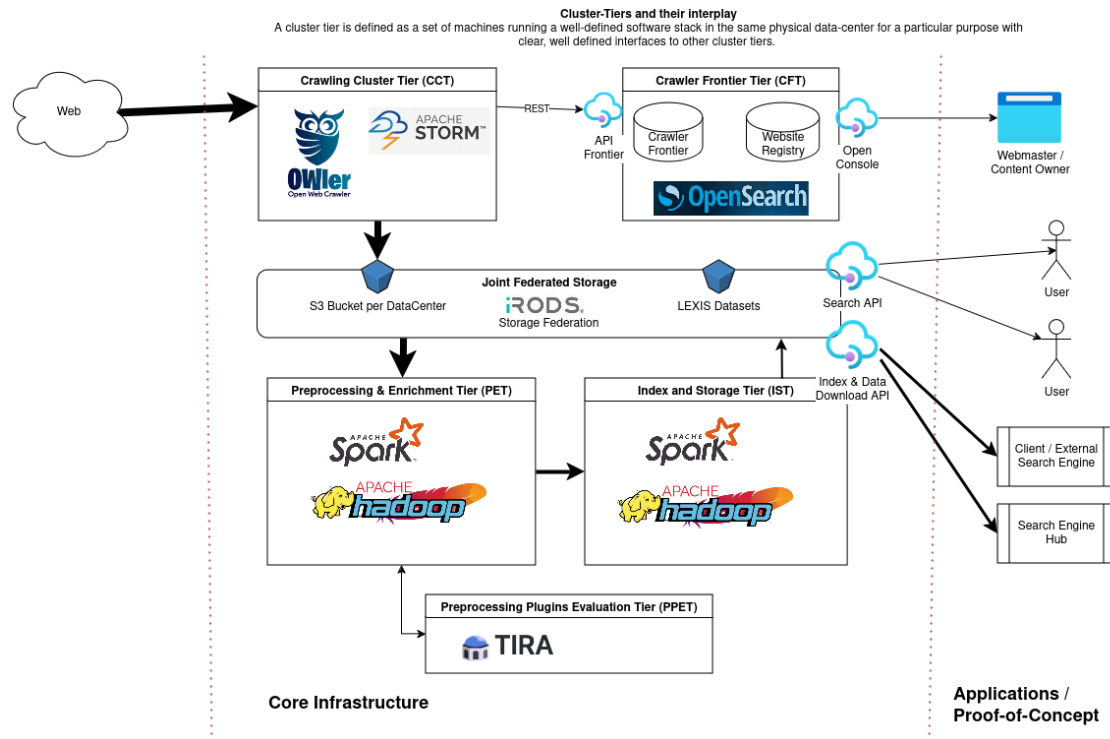
**Deployed at:** IT4I@VSB · BADW-LRZ · CSC · CERN + FSTP partners

## Scale (early 2026)

- **25 VMs** across 3–5 data centers running 24/7
- **100 M URLs / day** crawling · **~1.1 PB** raw data
- **~50 TB** index · **810+ public datasets**



# Cluster Tiers in Detail



## Crawling (WP 1)

- **OWler** (distributed StormCrawler): multi-site web crawl, WARC output
- Peak: ~100 M URLs/day at 4 partner sites (UNI PASSAU, LRZ, IT4I, CSC + EXAION)
- **ScyllaDB** frontier: dedup, politeness, cross-DC coordination

## Preprocessing (WP 2)

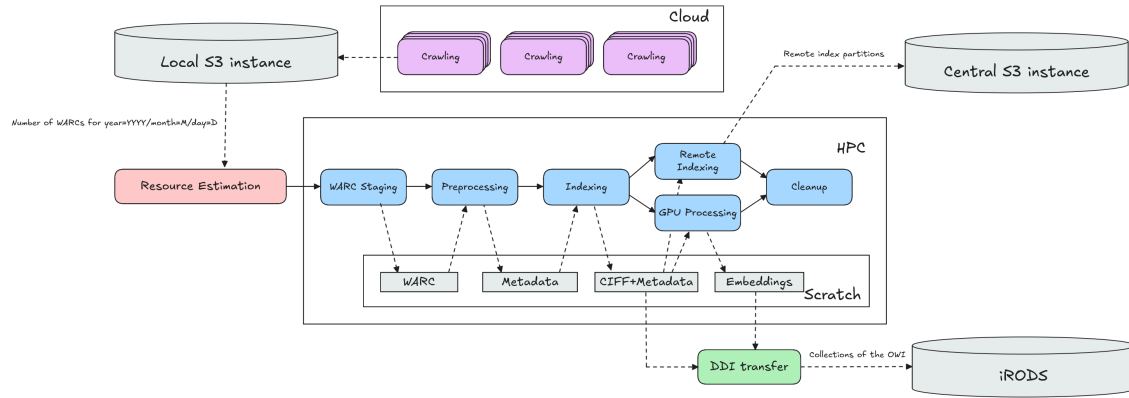
- **Resilipipe** on Spark: HTML → clean text, metadata extraction, topic/entity tagging
- **PPET** (singleton): plugin evaluation via TIRA before promotion to production
- GPU tier: quality-filtered docs → Jina V3 dense embeddings

## Indexing (WP 3)

- **Open Web Indexer** on Spark: Parquet → CIFF shards per collection + language

- Also produces **remote index** (DuckDB/OUR<sup>2</sup>S-queryable Parquet partitions)
- Federated storage: **iRODS** + S3 via LEXIS DDI layer

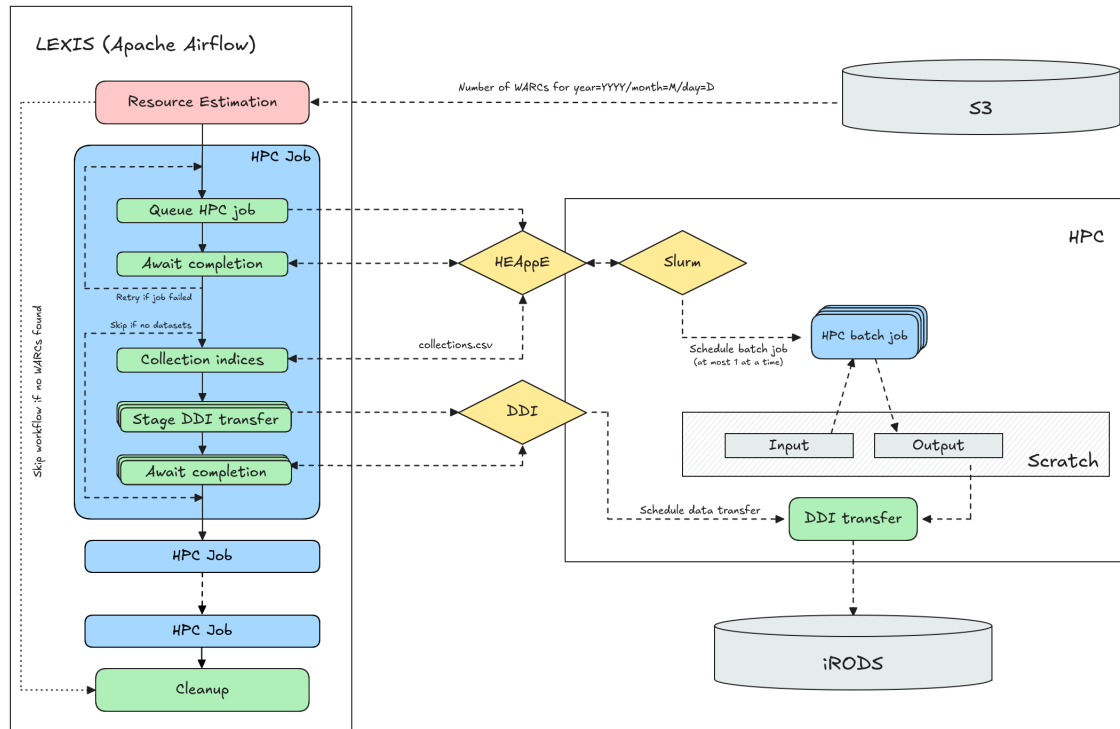
# Daily Workflow



- Each day at **03:00 AM** the pipeline triggers:
1. **WARC Staging** — copy new WARC files from S3 to HPC scratch
  2. **Resilipipe** — WARC → gzipped Parquet (text, metadata, enrichments)
  3. **Open Web Indexer** — Parquet → CIFF shards per collection / language
  4. **GPU Processing** — Resilipipe GPU → Jina V3 embeddings for quality slice
  5. **Remote Index** — Parquet partitions for DuckDB / OUR<sup>2</sup>S remote queries
  6. **DDI Transfer** — push output collections to iRODS via LEXIS DDI layer
  7. **Cleanup** — remove intermediate HPC scratch data

All intermediate data stored on **HPC scratch** between steps — restartable on partial failure.

# Workflow Orchestration via LEXIS



**LEXIS** (Apache Airflow) manages the full daily pipeline:

- **Resource estimation:** counts WARCs for that date → computes Slurm allocation
- **HEAppE:** submits and monitors HPC batch jobs at each data center
- **DDI layer:** transfers output collections to iRODS after each job succeeds
- Automatic retry on failure; session tokens refreshed throughout

**Workflow configuration** (JSON per DC):

- `crawl_location / hpc_location / storage_location` — Airflow Connections
- Per-task: `resources`, `max_retries`, `stage_ddi`, `depends_on`
- Dev / test / prod variants with reduced resource profiles

**Data center connections:** S3 (crawl), HEAppE (HPC scheduling), iRODS (persistent storage) — fully configurable per data center.

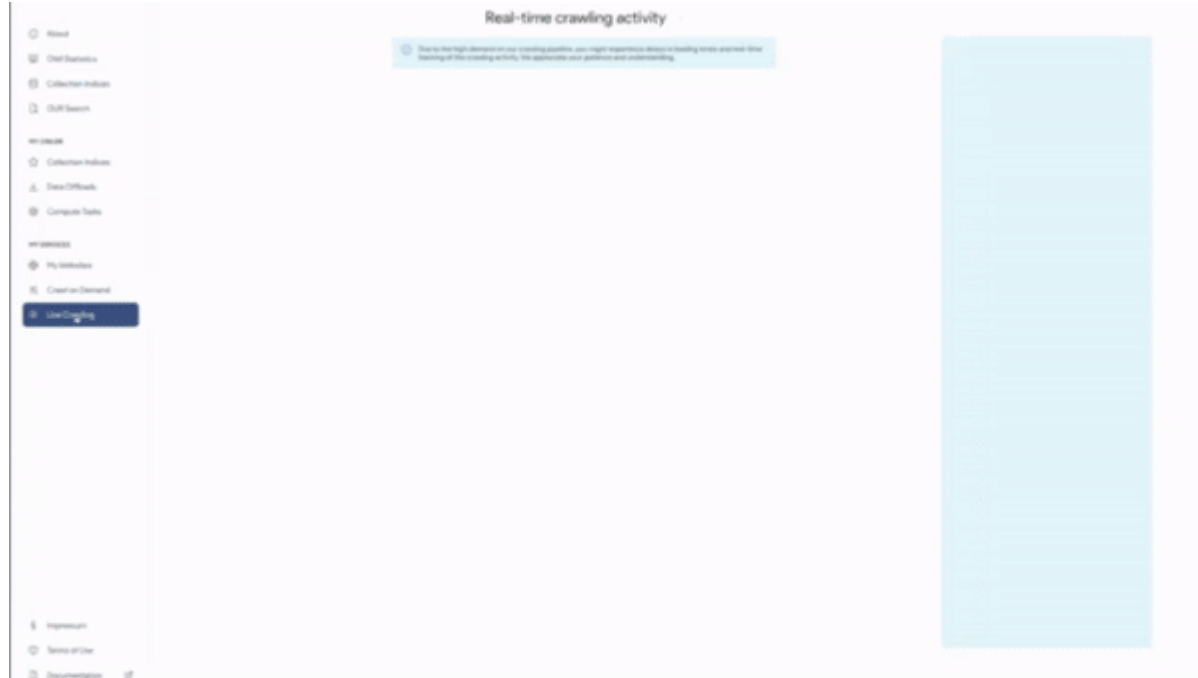
## Index & Federation

- ~**1.1 PB** raw data · ~**50 TB** index data · **810+ public datasets**
- **10–20 TB** new index data per month
- Approx. **factor 20** smaller than commercial indices (factor 10–50× for multimedia)

## Federated Architecture

- Distributed over **IT4I, CSC, LRZ, CERN** — onboarding further AI Factories
- Cross-DC workflow execution via LEXIS platform
- Federated storage: iRODS + S3; client-side SQL access via owilix / DuckDB
- Dense embeddings available on select slices

# Step 1: Crawling the Web



- 100 Million URLs/day across 25 machines in 3-4 data centers
- Central coordinator/queue at CSC/CERN, 2 billion URLs in queue
- Focus on European domains and European languages
- Respects robots.txt, Gen AI opt-out directives, no-index
- Filters potentially harmful and low-quality content



### All systems are online

All components are functioning properly. Data repositories and services are accessible.

Last Updated: 18/06/25, 06:15

| SYSTEMS |      | DESCRIPTION | COMPONENTS                       |                               |                              | STATUS |
|---------|------|-------------|----------------------------------|-------------------------------|------------------------------|--------|
| IT4I    | IT4I | connected   | <div><div></div>Repository</div> | <div><div></div>Project</div> | <div><div></div>Public</div> | ONLINE |
| lrz     | LRZ  | connected   | <div><div></div>Repository</div> | <div><div></div>Project</div> | <div><div></div>Public</div> | ONLINE |



Crawled Data  
1037.57 TB

+75.75TB  
last month

Jun 1, 2025  
last updated



WARC Datasets  
174

+23  
last month

Jun 1, 2025  
last updated



Public Datasets  
655

+165  
last month

N/A  
last updated

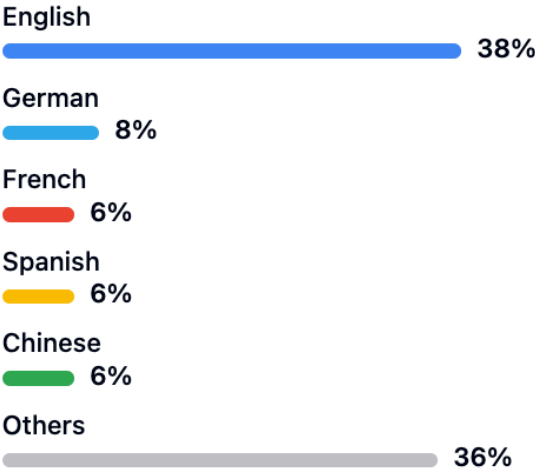


Open Web Index  
28.83 TB

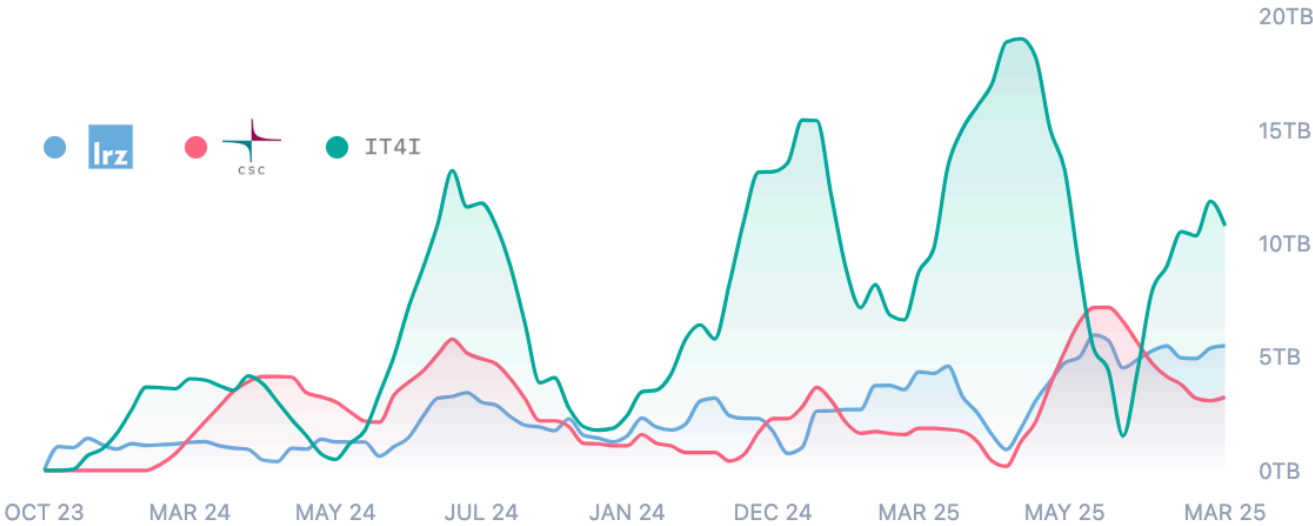
+2.1TB  
last month

N/A  
last updated

### Language Distribution ⓘ



### Crawl Volume by Datacenter ⓘ





# Step 2: Preprocessing

### Resilipipe

Resilipipe is an open source software framework that implements a scalable cluster-based web content analysis pipeline for web archive data based on [Resiliparse](#). It can be run on HPC clusters using [Apache Spark](#) and [Magpie](#). Users can expand the content analysis by implementing their own modules following our [interface](#).

Necessary configurations are made in the YAML files in [conf](#).

### Schema

The Parquet files produced by this pipeline will contain two sets of columns:

1. Fixed columns by the core part of the pipeline

2. Additional columns based on modules as outlined in the respective [README](#).

An overview over the columns for each schema version is given below:

► Schema Version 0.1.0

► Schema Version 0.1.1

► Schema Version 0.1.2

► Schema Version 0.2.0

► Schema Version 0.2.1

### Setup

Project information

19 Commits

1 Branch

0 Tags

648 KiB Project Storage

README

LICENSE

+ Add CHANGELOG

+ Add CONTRIBUTING

+ Enable Auto DevOps

+ Add Kubernetes cluster

+ Set up CI/CD

+ Add Wiki

+ Configure Integrations

Created on

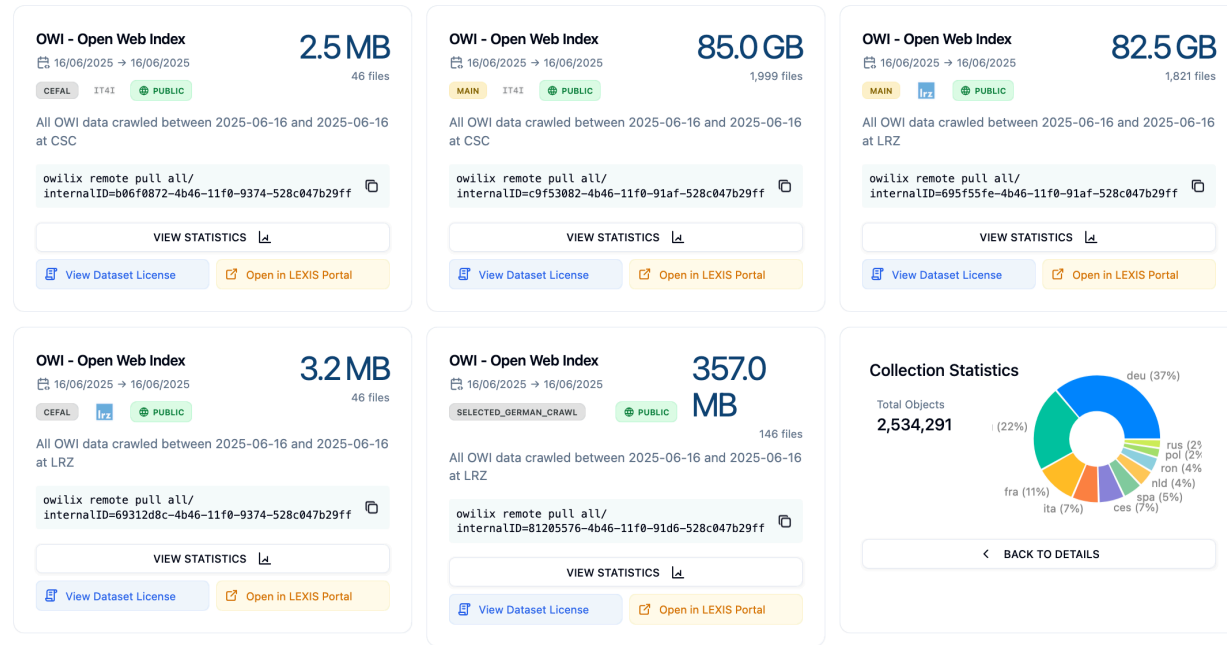
August 07, 2023

- Resilipipe

— open-source, modular, scalable pipeline
- HTML to cleaned text at scale (reduction ratio 1:5)
  - **Metadata extraction:** JSON-LD, Microdata, OpenGraph
  - **Semantic enrichment:** language, topics, geo-locations, NER
  - **Structured data extraction** from 60% of web pages: FAQ, opening hours, products, job postings
  - Open to community contributions via plugin architecture



# Step 3: Building and Distributing the Index



- One dataset per day per data center
- **Sparse full-text index:** CIFF format, SQL-based querying
- **Feature index:** link structure, linguistic style, genre, metadata
- **Dense index:** embeddings (in development), compression techniques for ANN search
- Pre-computed collection indices: curlie, legal, main, metadata/microdata
- Tooling for download and dataset management via Owilix

## OWI Dataset

- Year=2023

- month=12

- day=24

-language=eng

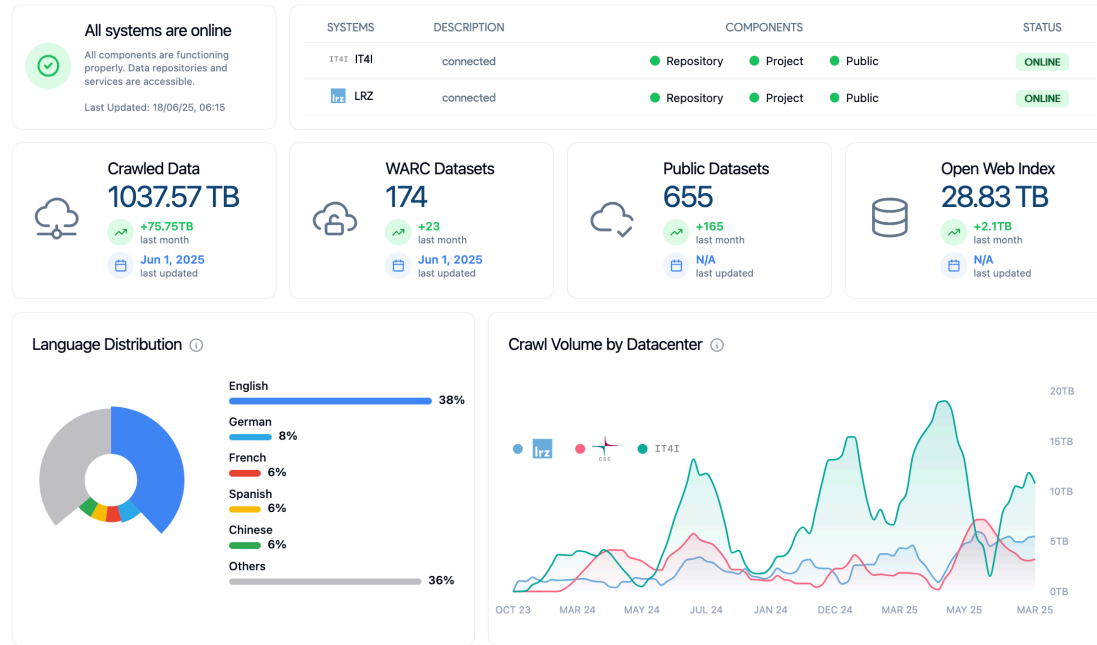
index.ciff.gz

metadata\_#.parquet

-language=....

# Accessing the Open Web Index

# Dashboard — Statistics & Datasets in Detail



## OWI Statistics Tab

- Repository health per data center (IT4I · LRZ)
- Key indicators: pipeline runs, HPC core-hours consumed, avg. duration, active pipelines
- Crawled data (1.3 PB), WARC datasets, public datasets, index size
- Language distribution chart + crawl volume timeline per DC
- Dataset collection distribution by resource type (main / GPU / curlie / embeddings)

## OWI Datasets Tab

- Filter: datacenter · format (WARC / Parquet / CIFF / Embeddings) · language · date range
- Infinite-scroll dataset cards: title, date, DC, public/private badge, file listing
- Parquet file preview in-browser; download via LEXIS Portal or owilix remote pull

# Dashboard — Corpora, Models & Search

## OWI Corpora

Curated, administrator-prepared collections for distribution:

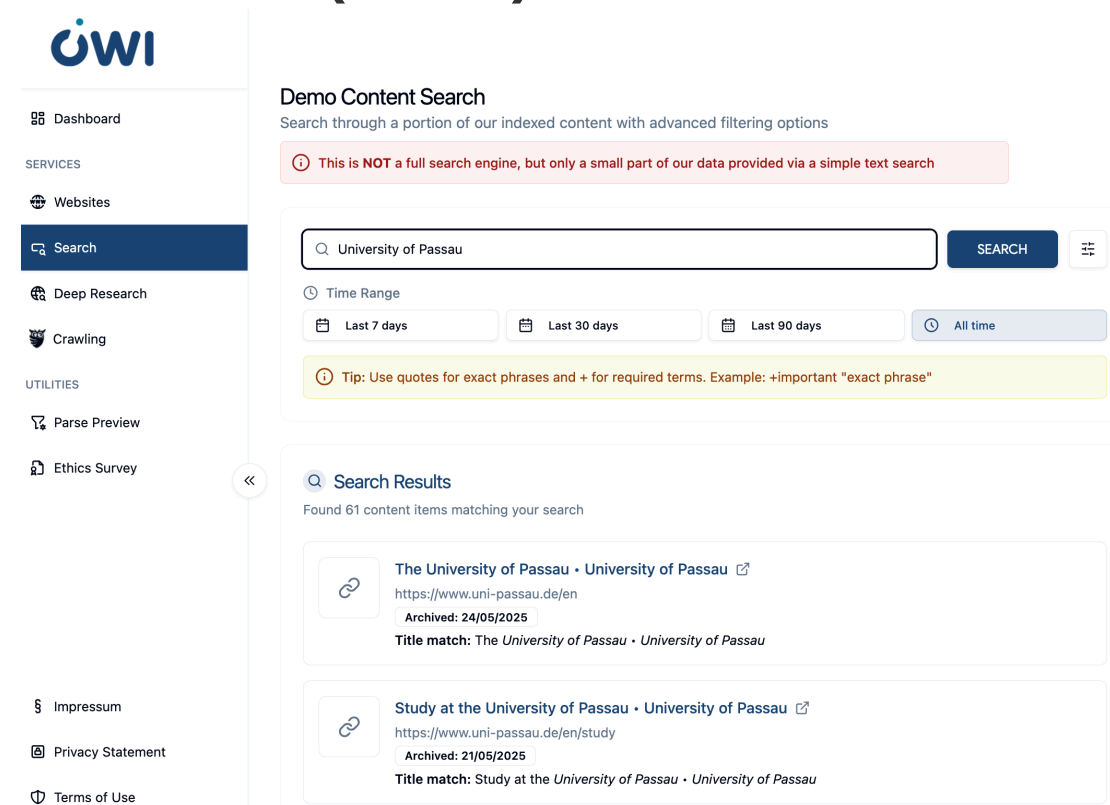
- Licensing metadata per corpus (creator, terms, full licence text)
- LEXIS integration for storage, access control and metadata
- Includes domain-specific and project-specific collections

## OWI Models

Registry of language models for web search:

- Browser-runnable models via WebAssembly (Wllama library)
- Cloud-based models (OpenAI, DeepSeek) for heavy tasks
- Model cards with training data, benchmarks,  and GGUF file listing

## OUR Search (Demo)



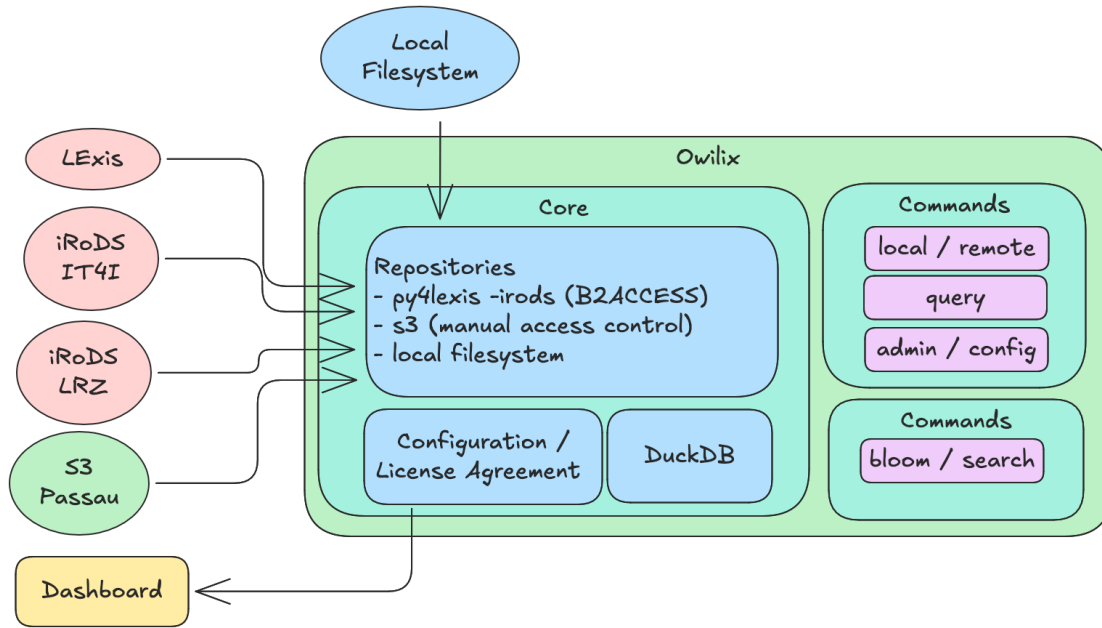
The screenshot shows the OWI Search Demo interface. On the left is a sidebar with the OWI logo and a navigation menu including Dashboard, Websites, Search (highlighted), Deep Research, Crawling, Parse Preview, and Ethics Survey. Below the menu are links for Impressum, Privacy Statement, and Terms of Use. The main content area is titled 'Demo Content Search' and includes a warning: 'This is NOT a full search engine, but only a small part of our data provided via a simple text search'. It features a search bar with 'University of Passau' entered, a 'SEARCH' button, and a 'Time Range' filter with options for 'Last 7 days', 'Last 30 days', 'Last 90 days', and 'All time'. A tip box suggests using quotes for exact phrases and '+' for required terms. The 'Search Results' section shows 'Found 61 content items matching your search' and displays two result cards. The first card is for 'The University of Passau' with a thumbnail, URL, archive date of 24/05/2025, and a title match. The second card is for 'Study at the University of Passau' with a similar layout and an archive date of 21/05/2025.

Simple + advanced field-specific search over a subset of the index; time-range filtering; result cards with thumbnail, title, URL

## Deep Research (Experimental)

AI-powered multi-step research: configurable depth + breadth, local or cloud LLM, interactive clarification mode

# owilix — The Open Web Index Client



Four main workflows:

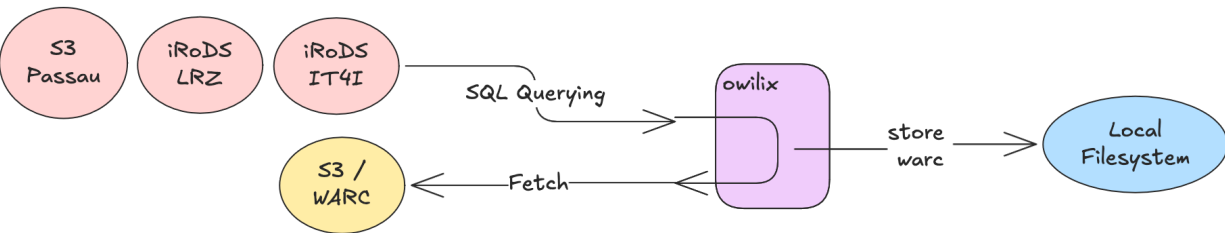
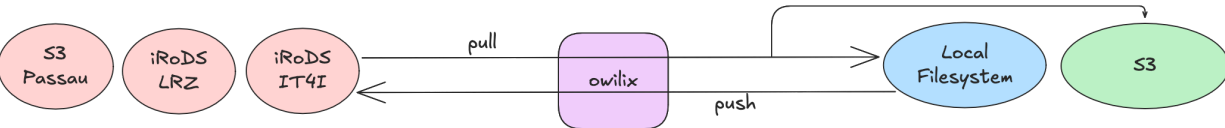
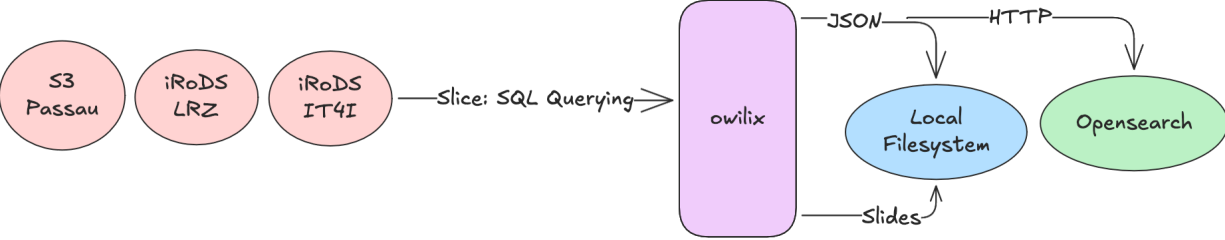
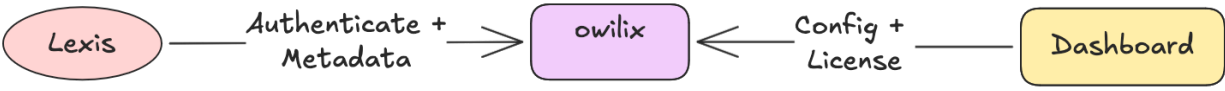
1. **Authentication** — B2ACCESS via LEXIS + Dashboard config (OpenID Connect / Keycloak)
2. **Sync Datasets** — pull/push between federated iRODS/S3 storage and local filesystem
3. **Query / Slice** — client-side SQL queries with JSON, HTTP, OpenSearch output
4. **WARC Access** — fetch original WARC files via SQL querying

Slice the index by:

- **Domain** (.de, .at, .fr, ...)
- **Topic** (Curlie.org hierarchy labels)
- **Language** (filename-based selection)
- **Sitelist** (CSV of URLs / domains / TLDs)
- **Structured data** / microdata type
- **Outlinks** for link graph analysis



# owilix Workflows



# Remote Index — Query Without Downloading

## What is the remote index?

A DuckDB / Apache Spark-queryable representation of the OWI — Parquet partitions aligned with CIFF shards, accessible from remote clients without downloading the full index.

## Example queries

```
-- Count documents per language
SELECT language, COUNT(*) as docs
FROM read_parquet('remote://lrz:latest/main/**')
GROUP BY language ORDER BY docs DESC;

-- Filter by domain
SELECT url, title FROM ... WHERE domain = 'wikipedia.org';
```

## Use cases

- Analytics and statistics on the live index
- Lightweight RAG pipeline ingestion
- Ad-hoc exploration before committing to a full  shard download

## OUR<sup>2</sup>S — Online Search API

OUR<sup>2</sup>S (built on Vespa) extends the remote index with a full REST search API:

- Structured document model: full page · snippets · embeddings
- Keyword, semantic and hybrid ranking modes
- Multi-index federation across data centers
- No local index needed — query directly over network

→ OUR<sup>2</sup>S details in the dedicated OUR<sup>2</sup>S column

## Access summary

| Mode   | Tool      | Authentication       |
|--------|-----------|----------------------|
| Browse | Dashboard | Anonymous / B2ACCESS |

| Mode                  | Tool                   | Authentication      |
|-----------------------|------------------------|---------------------|
| Download              | owilix                 | B2ACCESS<br>(LEXIS) |
| SQL queries           | DuckDB<br>remote       | B2ACCESS            |
| REST search           | OUR <sup>2</sup> S API | API key             |
| Conversational<br>RAG | MosaicRAG              | Open /<br>B2ACCESS  |

# MosaicRAG — Building RAG Systems on OWI

## From OWI Datasets to Conversational Search

MosaicRAG consumes OWI daily datasets directly — CIFF inverted index files, Parquet metadata records, and dense embeddings (published since Dec 2025) — and exposes them through a modular conversational interface.

### Pipeline for building your own MosaicRAG instance:

- 1. Download OWI slice with owilix (topic, domain, or language filter)
- 2. Ingest CIFF → inverted index; Parquet → DuckDB; embeddings → ChromaDB
- 3. Configure retrieval pipeline: sparse, dense, or hybrid
- 4. Choose reranker (TF-IDF · Embedding cross-encoder · Tournament LLM · Group LLM)
- 5. Optionally add summarization and relevance marking

Full documentation: [openwebindex.eu/book](https://openwebindex.eu/book) — "Building vertical search engines with MOSAIC"

### Supported LLMs

| Task              | Models                    |
|-------------------|---------------------------|
| Reranking         | gemma2, qwen2.5, llama3.1 |
| Summarization     | + DeepSeekv3              |
| Relevance marking | gemma2, qwen2.5, llama3.1 |

### Live Demo

Try conversational search on the Open Web Index at [mosaicrag.ows.eu](https://mosaicrag.ows.eu)

# Using the Open Web Index — Applications

## Realised Applications

- **Mosaic & MosaicRAG** — modular vertical search framework; science, health, arts demos → [mosaic.ows.eu](https://mosaic.ows.eu)
- **Earth Observation Search** — RAG QA + geo-contextualised search (DLR / Radboud)
- **Multilingual Scientific Index** — enriching Finland's Research.fi portal (CSC)
- **Restaurant Recommender** — privacy-preserving mobile app; 214-user pilot (A1 Slovenia)
- **Argumentation Search** — args.me extended with OWI pro/contra index → [args.me/owi](https://args.me/owi)

## Community Applications (FSTPs)

- **AKSE** — argumentation knowledge graphs
- **VERITAS / DEXAI** — verification-oriented RAG
- **OMMS** — enriching OpenStreetMap with POI metadata
- **DTCommerce** — SME e-commerce product enrichment
- **CIFFIL** — cross-collection index for municipal search
- **TILDE** — trustworthy health domain search

The OWI enables a full ecosystem of vertical search engines, RAG systems, and domain-specific applications

# MosaicRAG — Conversational Search on the OWI

**Modular RAG framework** extending MOSAIC with LLM-based capabilities and dense retrieval (WEBIS / TU Graz)

- **Two retrieval paths:** inverted index (CIFF) + dense retrieval
- **4 reranking strategies:** TF-IDF, Embedding cross-encoder, Tournament-style (LLM pairwise), Group-style (LLM window)
- **LLM backends:** self-hosted on-premise models for summarization
- **Conversational search:** follow-up questions about retrieval results; dual-window UI (search results + chat)
- **Workflow:** define retrieval pipeline → issue query → converse about results

## Key Features

- Relevance marking — LLM highlights relevant passages within documents
- Summarization — user-provided instructions applied per document
- Domain-specific indices: science, health, arts (8M+ docs)
- Fully open-source modular pipeline

## Data Pipeline

OWI daily datasets → CIFF (inverted index) + Parquet (metadata) + dense embeddings (since Dec 2025) → MosaicRAG



Search result list

Conversational search

Knowledge graph

Logs

Retrieval pipeline

Reset chat

Language model

gemma2

Text column

full-text

Add step

Reset everything

Get link

list universities in Austria

Here are some universities in Austria, categorized for clarity:

Public Universities:

- **University of Vienna (Universität Wien):** The largest and oldest university in Austria, renowned for its humanities, social sciences, and natural sciences programs.
- **Technical University of Vienna (Technische Universität Wien):** Focuses on engineering, technology, natural sciences, architecture, and mathematics.
- **University of Graz (Karl-Franzens-Universität Graz):** Known for its strong humanities, social sciences, law, medicine, and technical science faculties.
- **University of Innsbruck (Leopold-Franzens-Universität Innsbruck):** Offers programs in a wide range of disciplines, with strengths in the natural sciences, economics, and law.
- **Johannes Kepler University Linz (JKU Linz):** Specializes in engineering, natural sciences, social sciences, economics, and law.

MosaicDataSource

Output column name

full-text

MOSAIC service URL

https://mosaic.fe

Limit

20

Search index

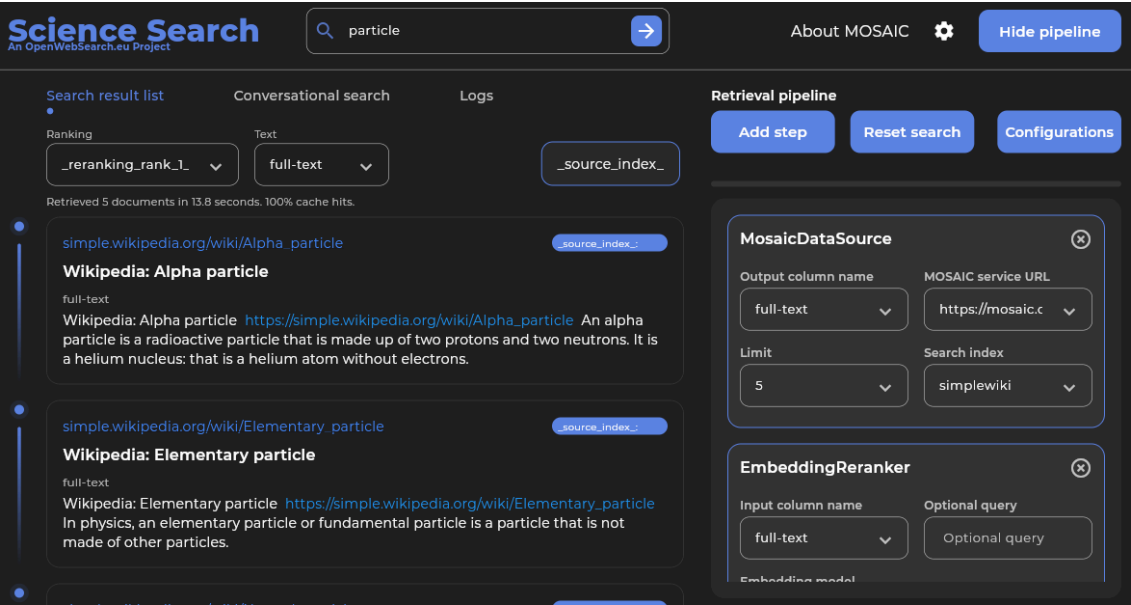
unis-austria

Demo: [mosaicrag.ows.eu](https://mosaicrag.ows.eu) · [mosaic.ows.eu](https://mosaic.ows.eu)

# Mosaic Search Applications · Earth Observation Search

## Mosaic Vertical Search (TU Graz)

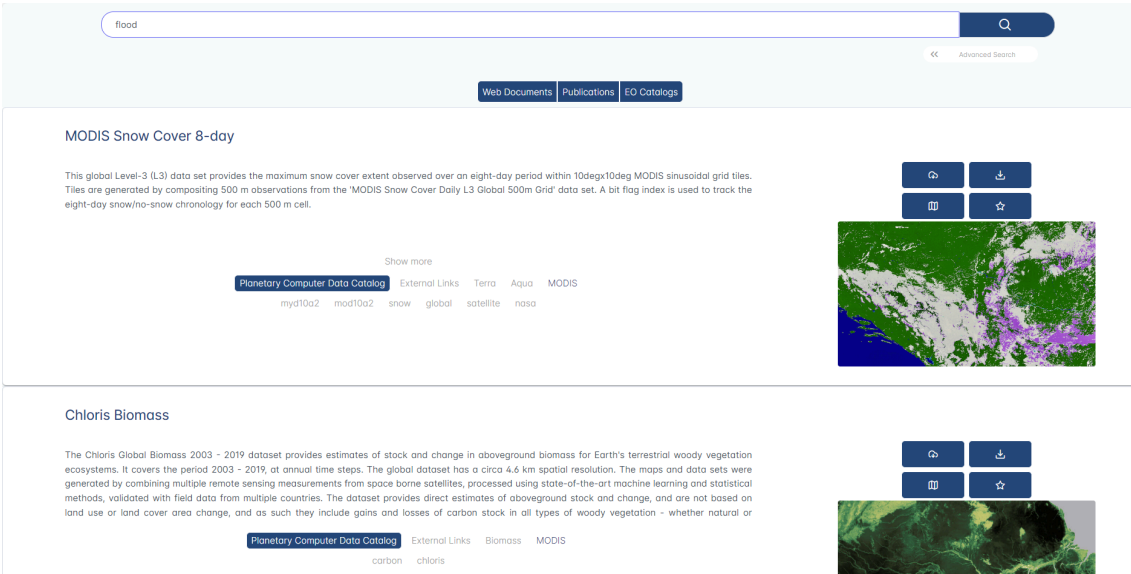
Modular framework for building vertical search engines on OWI slices — domain-specific indices for science, health, arts (8M+ docs)



Demo: [mosaic.ows.eu](https://mosaic.ows.eu) · [mosaic.ows.eu/demos](https://mosaic.ows.eu/demos)

## Earth Observation Search (DLR / Radboud)

RAG-based QA over scientific papers, datasets & web content; geo-contextualized multi-genre search on interactive maps; KG with 2M+ nodes (OpenAlex, PANGAEA, DLR Geoservice)



Demo: [eo-rag-demo.2.rahtiapp.fi](https://eo-rag-demo.2.rahtiapp.fi) · [ows-demo.2.rahtiapp.fi](https://ows-demo.2.rahtiapp.fi)



# Multilingual Scientific Index · Restaurant Recommender

## Multilingual Scientific Index (CSC)

Enriches Finland's Research.fi portal with OWI web data — Finnish & English science content classified by type (publication, blog, event); embedded directly in national research portal

### Energy-centric optimization of large-scale additive manufacturing deployment for sustainability through electrification

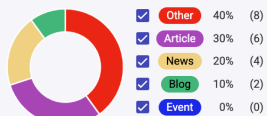
|                                     |                                                                       |                                                                                                                                                                                                                                                                                                                               |           |
|-------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Rahoitetun hankkeen kuvaus          |                                                                       | The employment of additive manufacturing (AM), commonly called 3D printing, in the industrial sector has been growing for many years. As the technology matures, it is expected that its use will become widespread and substitute other established forms of production. This project is...<br><a href="#">Näytä enemmän</a> |           |
| Aloitusvuosi                        | 2026                                                                  |                                                                                                                                                                                                                                                                                                                               |           |
| Päättymisvuosi                      | 2029                                                                  |                                                                                                                                                                                                                                                                                                                               |           |
| Myönnetty rahoitus                  | Humberto Almeida Junior                                               | Lappeenrannan–Lahden teknillinen yliopisto LUT                                                                                                                                                                                                                                                                                | 305 350 € |
| Rooli Suomen Akatemian konsortiossa | Partneri                                                              |                                                                                                                                                                                                                                                                                                                               |           |
| Muut osapuolet                      | Partneri                                                              | Aalto-yliopisto (372725)                                                                                                                                                                                                                                                                                                      | 298 731 € |
|                                     | Johtaja                                                               | Lappeenrannan–Lahden teknillinen yliopisto LUT (372723)                                                                                                                                                                                                                                                                       | 395 919 € |
| Rahoittaja                          | Suomen Akatemia                                                       |                                                                                                                                                                                                                                                                                                                               |           |
| Rahoitusmuoto                       | Suunnattu akatemiahanke                                               |                                                                                                                                                                                                                                                                                                                               |           |
| Haku                                | 2025 Tulevaisuuden kestävien energiaratkaisujen tutkimuksen haku 2025 |                                                                                                                                                                                                                                                                                                                               |           |
| Muut tiedot                         |                                                                       |                                                                                                                                                                                                                                                                                                                               |           |
| Rahoituspäätöksen numero            | 372724                                                                |                                                                                                                                                                                                                                                                                                                               |           |
| Tieteenalat                         | Kone- ja valmistustekniikka                                           |                                                                                                                                                                                                                                                                                                                               |           |
| Tutkimusalat                        | Kone- ja valmistustekniikka                                           |                                                                                                                                                                                                                                                                                                                               |           |

### Myöntöön liittyvä verkkosivu

Rahoitusmyöntöön ei ole tarjolla linkkejä tässä portaalissa

### Viittaukset muissa palveluissa

Viittauksia yhteensä: 20



### Hakutulokset 1 - 5

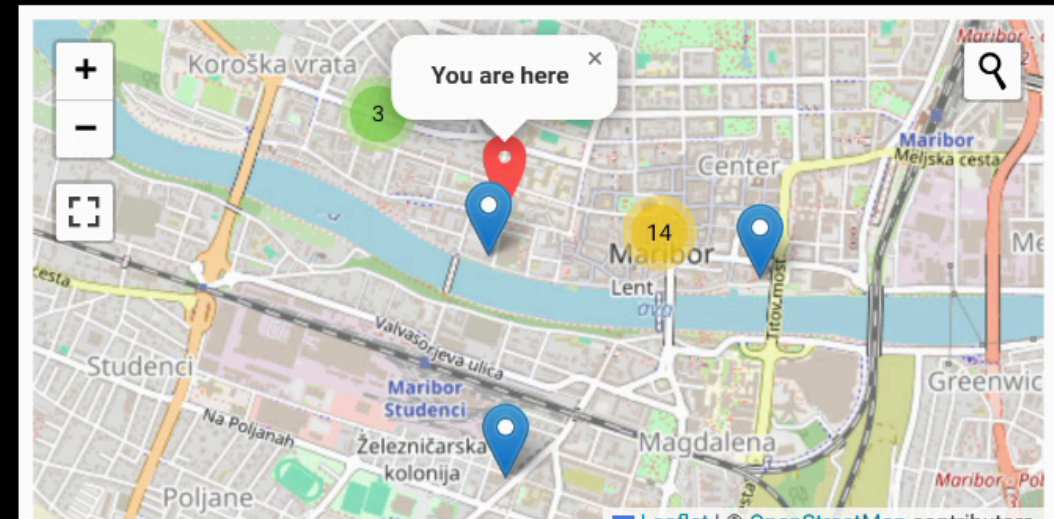
- <https://www.industryweek.com/...>  
3 Ways Additive Is Shaping the Factory of the Future | IndustryWeek
- <https://www.industryweek.com/...>  
These Industries are the Future of Additive Manufacturing | IndustryWeek
- <https://global-sci.com/...>  
Global Science Press: A Review of Process Optimization for Additive Manufacturing Based on Machine Learning
- <https://www.engineering.com/...>  
Customized Production Will Become Scalable with 3D Printing - Engineering.com
- <https://axial.acs.org/...>  
Welcome the Newest Associate Editors of ACS Publications Journals: Q3 2018 | ACS Publications Chemistry Blog

< 1 2 3 4 >

## Restaurant Recommender (A1 Slovenia)

Privacy-preserving mobile app; on-device personalization via query imputation; LLM-extracted restaurant features from OWI; 5-month pilot with 214 users — overall satisfaction 4.2 / 5

# Restaurant Recommender



Or search for restaurants in 1 km radius

Select City ▼

Bascarsija grill food



## Baščaršija

[Visit Restaurant Page](#)

Poshtna ulica 8, Maribor 2000 Slovenia 📞 +386 2 250 63 59

Cuisines: Barbecue; European; Grill; Slovenian; Eastern European Meals: Lunch; Dinner

Price: \$4 - \$13 Features: outdoor seating;

## Okrepcevalnica Sarajevo

[Visit Restaurant Page](#)

Gospodsvetska cesta 43c, Maribor 2000 Slovenia 📞 +386 2 252 34 31

Cuisines: Barbecue; European Meals: Lunch; Dinner Price: \$2 - \$11 Features: outdoor seating;

# Argumentation Search · CERN Search Applications

## Argumentation Search (WEBIS)

args.me extended with OWI: pro & contra arguments on 31 controversial topics; argument quality scoring, key point matching, temporal trend analysis

## CERN Search Applications

- **Institutional**: 25K+ pages crawled from CERN internal web; full OWS pipeline → MOSAIC index for AccGPT knowledge base
- **Nooon**: disability knowledge search (2.59M docs) for HR & Diversity & Inclusion



args

We should abolish the right to keep and bear arms

violence

All

Search

520 results

46 ms

Pro-Arguments

Con-Arguments

ranking score: 9.051

source

Argumentative Segment

Since the start of the year, there have been 80 mass shootings and more than 3,300 deaths due to gun **violence** nationwide, according to the Gun **Violence** Archive. We don't need any more thoughts and prayers from elected leaders. We shouldn't have any more moments of **silence** or vigils. We need action.

Key Point

Quality Score: 0.880

Gun ownership allows for mass-shootings/general gun violence

ranking score: 8.636

source

Argumentative Segment

"Any successful strategy to reduce gun **violence** requires preventing the diversion of lawful firearms into unlawful commerce," Steven Dettelbach said at the time. "Once there, these firearms end up in the hands of people who are sometimes **violent** criminals and intend to do harm to the people with whom we live, the innocent people who are victims and survivors of gun **violence**."

Key Point

Quality Score: -0.690

Guns can fall into the wrong hands

ranking score: 11.986

source

Argumentative Segment

Councilman Johnson, repeat after me: **Violence** is **Violence**. The number one cause of **violence** is **violent** people. While semantically there is an argument that some people aren't **violent** until provoked, drunk and/or stoned, exposed to constant **violence** in entertainment media, video games etc., the fact remains that **violence** always, 100% of the time, involves at least one **violent** person. Blaming guns for **violence** is like blaming hammers for shoddy home construction.

Key Point

Quality Score: 0.660

ranking score: 8.489

source

Argumentative Segment

No matter how many times actual, real-life case studies are laid out for Cleveland's elected officials, they refuse to recognize that the only PROVEN method of reducing **violence** is removing **violent** persons from circulation, either through imprisonment or by vibrantly empowering the potential victims to remove the **violent** person when confronted by one.

Key Point

Quality Score: 0.870

Gun ownership promotes self protection

Demo: [args.me/owi](https://args.me/owi) · [args.me](https://args.me)

MOSAIC

ows-lb-2.cern.ch/ui/

80%

MOSAIC Search

Search term: disability

Search

Index Info

Geo Filter:

West:

East:

North:

South:

Index:

Language:

Limit:

Keyword:

☐ default / all

☒ disability-index

☒ default / all

☐ English

☐ German

☒ default / 20

☐ 10 items

☐ 50 items

☐ 1,000,000

Search URL: https://ows-lb-2.cern.ch/mosaic/search?q=disability&index=disability-index

Search result for term: "disability"

Index: disability-index

Number of items: 20

Cockrell Hill Social Security Disability Attorneys | OneCLE Texas Attorney Directory

Kraft Dallas, TX Social Security Disability Lawyer with 54 years of experience (214) 999-9999 2777 N Stemmons Fwy Suite 1300 Dallas, TX 75207 Free ConsultationSocial Security Disability and Personal Injury Baylor Law School Show Preview View Website

Metadata: language:eng, word count:4273, index date:2025-08-10 00:13

Locations:

Keywords: Attorney • Attorneys • Social Security Disability • Cockrell Hill • Texas • https://lawyers.onecle.com/lawyers/social-security-disability/texas/cockrell-hill Insurance Coordinator Manual

Partial disability must result from the same condition as the total disability. Proof of partial disability must be submitted within 31 days of the date the employee's total disability period ends. Pa

Metadata: language:eng, word count:20858, index date:2025-08-10 00:51

Locations:

Keywords: https://oklahoma.gov/egid/insurance-coordinators/manual.html

House of Lords/Commons - Joint Committee on the Draft Disability Discrimination Bill - First Report

Disability Rights Commission, DDB 1, para 10.6.1. British Council of Disabled People, DDB 6, para 9. Law Society, DDB 15, p7. 15(a): The Government agrees with the recommendations of the DRTF that MPs

Metadata: language:eng, word count:12308, index date:2025-08-10 00:25

Locations:

Keywords: https://publications.parliament.uk/pa/jt200304/jtselect/jtdisab/82/8222.htm

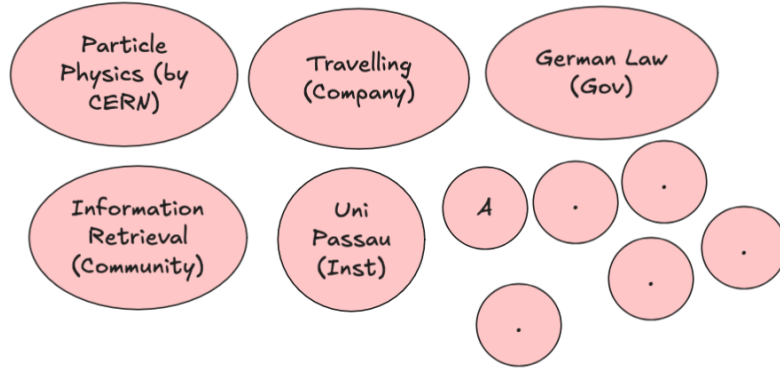
UKRI diversity data for funding applicants and awardees 2020 to 21 update – UKRI

After removing this opportunity, the award rate for fellows reporting a disability was 22% compared with

## Third-Party Projects (FSTPs)

- **AKSE** — argumentation knowledge graph for structured deliberation
- **VERITAS / DEXAI** — verification-oriented RAG for sensitive geopolitical contexts (Ukraine)
- **OMMS** — enriching OpenStreetMap POI with OWI structured data
- **DTCommerce** — product extraction & enrichment pipeline for SME e-commerce
- **CIFFIL** — cross-collection index interoperability for municipal search (Dutch pilots)
- **TILDE** — trustworthy health search with bias detection and entity-level retrieval

# Towards Federated RAG Systems



- RAG / Agentic systems require focused, high-quality collections
- Web information redundancy increases due to LLMs — no one-size-fits-all retrieval
- OpenWebSearch.eu supports low-cost creation of topic-specific RAG systems

## Key ideas:

- Retrieve from well-designed RAG systems instead of generic search engines
- Topic-specific, deduplicated and curated content
- Domain-optimized ranking, aggregation and summarization
- Lightweight browser-based federation with personal data integration

**OwlerLite** (WWW 2026) — privacy-respecting browser extension for scope- and freshness-aware retrieval: user-defined **scopes** (trusted sources), semantic

freshness detection via SimHash + vector similarity, Scope Fidelity improved from 0.64 → **0.83** (negligible NDCG loss)



# Using the Open Web Index — Data & Analytics

## Published Datasets

| Dataset         | Scale                 | Use case                        |
|-----------------|-----------------------|---------------------------------|
| WebFAQ          | 96M Q&A, 75 langs     | QA training, cross-lingual IR   |
| WebFAQ 2.0      | 198M pairs, 108 langs | Dense retrieval, hard negatives |
| Imprints        | 5.25M hosts, geocoded | Enterprise analytics, Eurostat  |
| German Commons  | 154B tokens           | German LLM pre-training         |
| OWS-Curlie-2025 | 20.7M docs + qrels    | IR evaluation benchmark         |

All datasets available at [openwebindex.eu/corpora](https://openwebindex.eu/corpora)

## Analytics Potential

- **Official Statistics:** enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring:** supply chain analysis, news monitoring (JRC)
- **AI Training Data:** domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks:** web-scale IR test collections with relevance assessments
- **Trend Analysis:** temporal web monitoring at Petabyte scale

The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem



# WebFAQ — Multilingual FAQ from the Web

- **96 million** Q&A pairs across **75+ languages**
- Extracted from structured FAQ markup (JSON-LD, Microdata) in the Open Web Index
- Fields: question, answer, URL, origin, topic, question type
- Parquet format, hosted on Hugging Face

**Use cases:** multilingual QA training, information retrieval, cross-lingual transfer learning

[huggingface.co/datasets/PaDaS-Lab/webfaq](https://huggingface.co/datasets/PaDaS-Lab/webfaq)

| Top languages | Pairs     |
|---------------|-----------|
| English       | ~30M      |
| German        | ~8M       |
| French        | ~6M       |
| Spanish       | ~5M       |
| ...           | 75+ langs |

# WebFAQ 2.0 — Hard Negatives for Dense Retrieval

- **198 million** FAQ-based Q&A pairs across **108 languages**
- **14.3 million** bilingual aligned QA pairs — largest FAQ-based resource
- **1.25 million** queries with mined hard negatives (20 languages)
- 200 cross-encoder scored negatives per query

## Key contributions:

1. Novel web crawling for diverse content extraction
2. Hard negatives for dense retriever training
3. Contrastive Learning & Knowledge Distillation
4. Open datasets and training scripts

---

*Dinzinger, Caspari, Salman, Topi, Mitrović & Granitzer. "WebFAQ 2.0: A Multilingual QA Dataset with Mined Hard Negatives for Dense Retrieval." arXiv:2602.17327, 2025.*

# Imprints Dataset — Geospatially Grounded Web Curation

- **5.25 million** hosts, **47.8 GB**
- Each website annotated along two dimensions:
  - **Topical classification** via LLM (Foursquare category hierarchy)
  - **Geographic grounding** from legal imprints via geocoding
- **3.08 million** websites geocoded from extracted addresses
- Page types: main page, imprint, contact, about us, privacy, terms

| Country     | Geocoded  | Share |
|-------------|-----------|-------|
| Germany     | 2,636,913 | 85.5% |
| Austria     | 223,805   | 7.3%  |
| Switzerland | 160,683   | 5.2%  |

| Top-Level Topic         | Share |
|-------------------------|-------|
| Business & Professional | 41.6% |
| Community & Government  | 16.2% |
| Arts & Entertainment    | 12.2% |



# German Commons — 154B Tokens for German LLMs

- 154.56 billion tokens, 35.78 million documents
- Aggregated from 41 diverse sources under open licenses
- Quality-filtered, deduplicated, OCR scored

| Domain             | Tokens |
|--------------------|--------|
| News Commons       | 72.67B |
| Cultural Commons   | 54.49B |
| Web Commons        | 19.89B |
| Political Commons  | 3.57B  |
| Legal Commons      | 2.99B  |
| Scientific Commons | 0.84B  |
| Economics Commons  | 0.11B  |

Gienapp, Schröder, Schweter et al. "The German Commons — 154 Billion Tokens of Openly Licensed Text for German Language Models." arXiv:2510.13996, 2025.

# OWS-Curlie-2025 — Web Search Test Collection

- **20.7 million** documents from 6 months of OWI crawl data (March–August 2025)
- English documents from the Curlie subcollection (websites labeled in the Curlie directory)
- Three evaluation splits with student-provided topics and relevance assessments
- Jina V3 embeddings, ClFF indexes for sparse retrieval
- Accessible via owilix CLI and ir-datasets-owi Python package

| Split       | Documents  | Embeddings | Qrels |
|-------------|------------|------------|-------|
| all         | 20,746,181 | --         | --    |
| radboud-val | 63,639     | yes        | yes   |
| radboud     | 90,852     | yes        | --    |
| jena-kassel | 77,382     | yes        | --    |

[openwebindex.eu/corpora](https://openwebindex.eu/corpora) · License: OWIL v1.0



# Ethical, Legal & Societal Aspects (ELSA)

# ELSA Framework — Values, Risks & Mitigation

## 10 Ethical Values

1. Transparency
2. Safety & Security
3. Openness
4. Responsibility
5. Privacy
6. Justice
7. Trustworthiness
8. Accountability
9. Sustainability
10. Sovereignty / Autonomy

## 5 Ethical Risks

1. Control of indexed websites & take-down requests — censorship vs. coverage
2. Flawed content in the OWI — false, misleading, offensive material
3. Inaccurate metadata — errors in ingestion or preprocessing
4. Misuse by third parties — privacy violations, unethical applications
5. Unsuitable governance — undemocratic control, security breaches

## ELSA Deliverables

Three iterations (D6.1 → D6.2 → D6.3), final version subsumes all.  
Developed with:

- 3 multi-day ethics workshops (CAIS / Stiftung Mercator)
- 3 legal FSTP projects (LISA, LOREN, ALMASTIC)
- OSF Working Groups on Ethics & Law

## 6 Stakeholder Groups

End users · Index users ·  
Developers · Researchers &  
institutions · Content providers ·  
Governance

# Legal Challenges & Regulatory Landscape

## Regulatory Framework

- **8+ EU regulations** apply: GDPR, Copyright Directive, DSM Directive, DSA, DMA, Data Act, Data Governance Act, AI Act
- **Copyright is the biggest challenge** — impacts crawling, redistribution, and LLM training in Europe
- No guiding court decisions yet — first time an open web index operates in the EU
- DSA classification unclear — OWI does not fit neatly as hosting service, platform, or search engine
- **AI Act:** OWI classified as "**limited risk**" — research exemption applies during project; post-project needs reassessment

## Legal FSTP Projects (published on Zenodo)

- **LISA** — Legal Framework for OWI (Matthias Wendland)
- **LOREN** — Legal Open European Web Index (Paul Johannes, Maxi Nebel)
- **ALMASTIC** — Assessing Legal Risks (Kai Erenli, Andreas Gruber)

## Key Legal Risks

- Personal data in crawled content (GDPR Art. 6(1)(f) legitimate interest)
- Copyright infringement — images, paywalled articles, plain text
- Liability as digital intermediary
- Illegal content obligations
- LLM training restrictions (no\_genai flag)



# 10 Technical & Organisational Mitigation Measures

| #   | Measure                         | Key Mechanism                                                        |
|-----|---------------------------------|----------------------------------------------------------------------|
| M1  | Website Management              | robots.txt opt-out, blacklisting, quality prioritisation             |
| M2  | Take-down Management            | Dashboard submission, logged removal, blacklist propagation          |
| M3  | Content Analysis                | Quality scoring, trigger warnings, ad detection, plugin architecture |
| M4  | Access Control                  | B2ACCESS registration, immediate blocking of misuse                  |
| M5  | Transparency Guideline          | 27-aspect guideline for third-party search apps                      |
| M6  | Data Protection & Privacy       | Anonymised queries, minimal user data, reusable model                |
| M7  | Safety & Security               | Regular security meetings, incident reporting                        |
| M8  | OWI Licence                     | Research (default) · Commercial (OWICL) · Data Access (raw WARC)     |
| M9  | Legal & Ethical Self-Assessment | Questionnaire at download time, ethics score, signed PDF certificate |
| M10 | Ethical & Legal Advice          | Ethics workshops, legal FSTPs, OSF working groups                    |

## Three Licence Types

- **Research** — default; included in owilix
- **Commercial (OWICL)** — requires personal contact with coordinator
- **Data Access** — raw WARC; research only, personal request

### Open Web Index Licence (OWIL) – Version 1.0

Last Updated: May 8th, 2024

#### Preamble

To promote an open human-centred search engine market and provide a solid basis for building front ends for offering a true choice to internet users in selection of preferable search engines, OpenWebSearch.EU develops and pilots the core of a European Open Web Index (OWI). It is the foundation for an open and extensible European Open Web Search and Analysis Infrastructure.

#### Definitions

“Index Data” – The OpenWebSearch.eu Index Data consists of processed web data derived from previously crawled web data. They are processed in a way that they serve as basis for search applications and web search engines. Hence, they consist of a search index (list of terms and related web documents) and meta data of the original web documents including the plain texts and processed information. The Index Data is distributed over several data centres and organised in partitions.

# Open Webmaster Console — Transparency Tools

## Website Management

### Website Management

Explore indexed website data, verify ownership to gain management access, and submit content removal requests

SEARCH

ⓘ

Note: Search results reflect our database as of December 08, 2025. We're continuously updating our index to provide the most current information.

📈 Recently Indexed Websites

Browse recently discovered websites in our database (225 total)

3zza0.xn163.com

View Details →

qroq6.szsiqu.com

View Details →

06npb.xn163.com

View Details →

maruhide.com

View Details →

gcmu2.jimforl.com

View Details →

shop1385571536059.qiyeku.cn

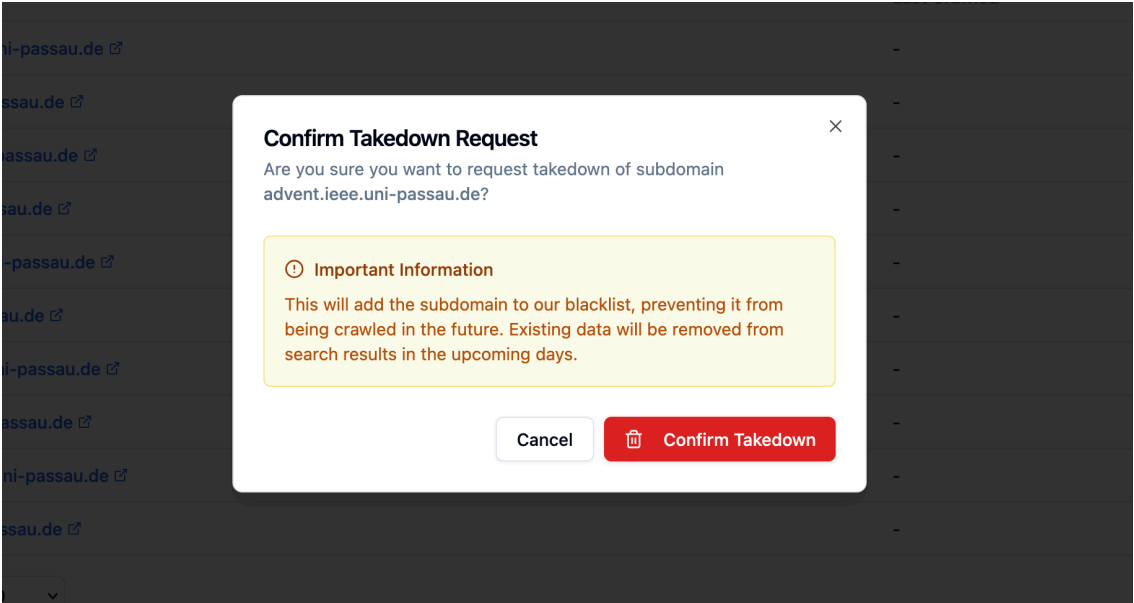
View Details →

zjg.haoyun56.com

View Details →

- Search any domain in the OWI — crawl stats, content types, language distribution,

## Content Removal & Rights



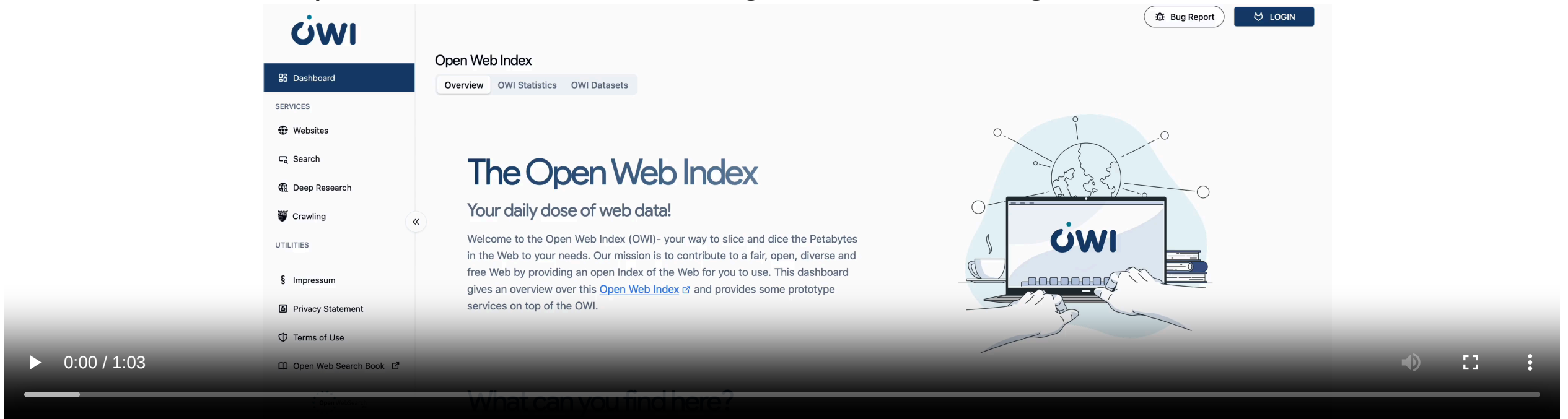
- Verified domain owners can submit **takedown requests** — subdomain or URL level
- Removed content is added to a blacklist; changelog propagated to all index users
- **Ethical self-assessment** survey required before enabling dataset downloads
- **OWLer crawler config**: express crawl constraints directly in the dashboard

subdomains

- **Domain verification:** prove ownership to unlock management capabilities

# Transparency & Webmaster Engagement

- Dashboard provides full transparency on crawling activity
- Webmasters can claim ownership of their sites
- Express usage constraints and issue takedown requests
- Gen AI exclusion protocols: robots.txt, user-agent, X-Robots-Tag



# FreeWebSearch Charter & Regulatory Reform

**#FreeWebSearch Charter** — launched 29 September 2025

10 Principles: transparency & explainability · competition & independent infrastructure · privacy without tracking · user/creator/advertiser control · equal access & non-discrimination · information diversity · environmental & social responsibility · integrity against disinformation · search competences & critical awareness · democratic control across borders

Four commitments: (a) serves society, people first; (b) respects democratic values; (c) transparent and open; (d) reflects internet diversity

**81 signatories** as of February 2026 — [openwebsearch.eu/charter](https://openwebsearch.eu/charter)

## 3 Regulatory Reform Proposals

1. **Copyright** — new statutory exception for open web indexes in DSM Directive
2. **Data Protection** — public-interest exemption under Art. 6(1)(e) GDPR
3. **Interoperability** — classify OWI as neutral data intermediary under DSA

Promoted via EU Data Union Strategy consultation (July 2025) and European Parliament breakfast event (December 2025)

# Societal Impact

- **Digital sovereignty** — reducing dependency on non-European search engines and AI data pipelines
- **Environmental sustainability** — reducing redundant crawling; shared infrastructure; data centre energy efficiency (CERN heat reuse)
- **Economic opportunity** — ecosystem enabling SMEs and startups; market potential study shows **ROI within 4 years**, projected **net benefit of ~4.5 B EUR over a decade**
- **Democracy & inclusiveness** — unbiased information access for informed citizens; multilingual/multicultural coverage; digital literacy

## Proposal: European Web Data Clearing House

A public infrastructure that collects, preprocesses and serves bulk web data as a certified, legally compliant data source — eliminating liability risks for European innovators at scale



# In Summary

## What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

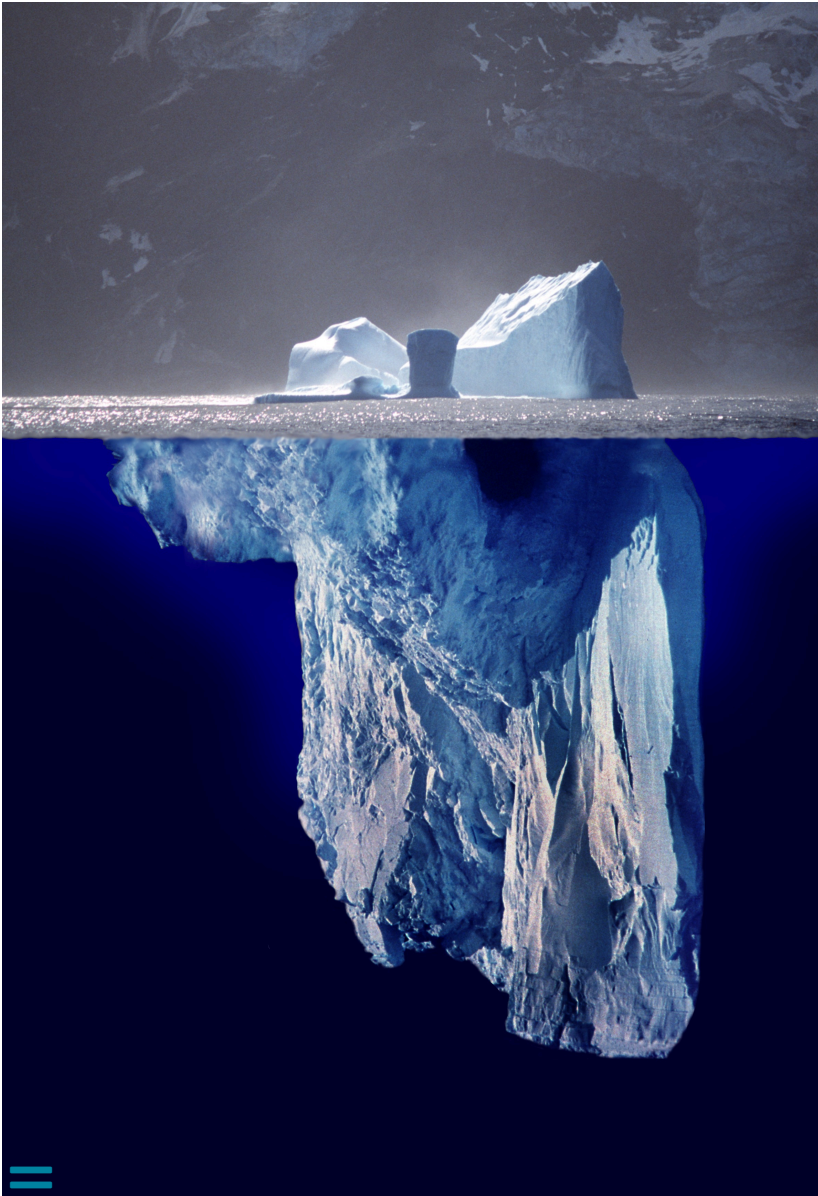
## Why it matters

- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable



# Key Messages



- Web-data and web search are **critical for Europe's digital sovereignty**
- **Reducing entry barriers** for researchers, innovators, and businesses
- Opening the core element in web search: the **Open Web Index**
- Open software stack, open pipelines, open data
- Distributed, data-centric, web-scale compute infrastructure
- Web data for **AI and Retrieval Augmented Generation**
- Building an **ecosystem** around data, software, and infrastructure
- Transparency and control for legal, ethical, and societal values

## We are looking for ...

- **Researchers & tech innovators** — develop new search & retrieval paradigms and content analysis algorithms
- **Data centres** — help hosting a distributed Open Web Index
- **Industry & business partners** — discover the business models of an Open Web Index
- **Policy makers** — help shaping the governance of an open search ecosystem

## Contact

- Email: [join@openwebsearch.org](mailto:join@openwebsearch.org)
- Community: [community@openwebsearch.eu](mailto:community@openwebsearch.eu)
- Website: [openwebsearch.eu](https://openwebsearch.eu)
- Dashboard: [openwebindex.eu](https://openwebindex.eu)
- Mattermost: [openwebsearch.eu/community](https://openwebsearch.eu/community)

**Thanks. Questions? Comments? Feedback?**



# Before we come to the Questions

# We are looking for ...

## Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

## Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

## Get in touch

|                 |                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
|                 |                                                                                                                 |
| Website         | <a href="https://openwebsearch.eu">openwebsearch.eu</a>                                                         |
| Dashboard       | <a href="https://openwebindex.eu">openwebindex.eu</a>                                                           |
| Code Repository | <a href="https://opencode.it4i.eu/openwebsearcheu-public/">https://opencode.it4i.eu/openwebsearcheu-public/</a> |
| Community       | <a href="https://openwebsearch.eu/community">openwebsearch.eu/community</a>                                     |
| Join            | <a href="mailto:join@openwebsearch.org">join@openwebsearch.org</a>                                              |
| General         | <a href="mailto:community@openwebsearch.eu">community@openwebsearch.eu</a>                                      |

Thank you — Questions, Comments, Feedback very welcome!